



Development of a Device for Non-Invasive Treatment of Alzheimer's using Electric Field



Master's Thesis Project

Presented by:

K S Ranjith Kumar

(Roll No.: 244159016)

Under the supervision of

Prof Harshal B Nemade

Jyoti and Bhupat Mehta School of Health Sciences and Technology

Indian Institute of Technology Guwahati

Guwahati, Assam 781039, India

11th June 2026

Motivation

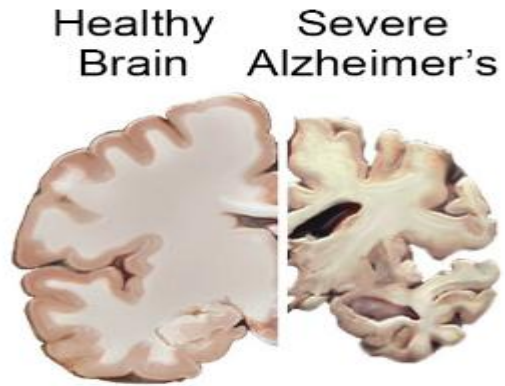


Figure 1. Comparison between normal brain and patient suffering from Alzheimer's Disease

(Adapted from National Institute on Aging, NIH)

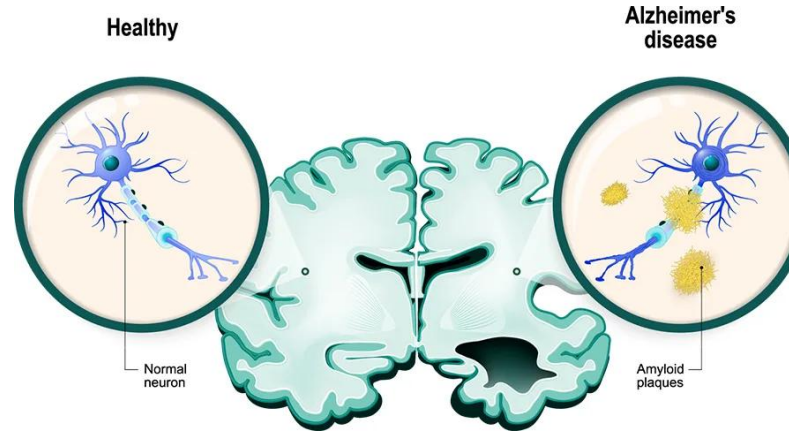


Figure 2. Presence of Amyloid beta plaques in patient suffering from Alzheimer's Disease

(Adapted from National Institute on Aging, 2024 & Research outreach, 2023)

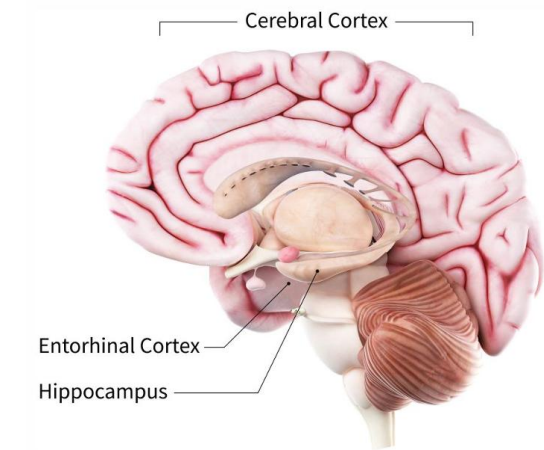


Figure 3. Depicting regions of brain affected by Alzheimer's

(Adapted from Springer Nature, 2024)

- Alzheimer's disease destroys neurons and their connection in parts of the brain involved in memory, including the entorhinal cortex and hippocampus.
- It later affects areas in the cerebral cortex responsible for language, reasoning and social behavior, over a time person with AD gradually loses his or her ability to live or function independently. Ultimately disease is fatal.
- 55M+ people affected globally
- Caused by β -Amyloid Plaques + Tau tangles $\rightarrow$ Neuronal Death
- Current Therapies:
 - Drugs = Costly and have limited effect, Existing Simulation Methods mainly target mainly target symptoms and cognition (alter neuronal firing)
 - Implants = Invasive, Risky
- Unmet Need : Safe, affordable, non-invasive intervention

Scientific Basis

Electric Field Disruption of Amyloid Aggregation: Potential Noninvasive Therapy for Alzheimer's Disease

Jahnu Saikia,[†] Gaurav Pandey,[†] Sajitha Sasidharan,[†] Ferrin Antony,[†] Harshal B. Nemade,[‡] Sachin Kumar,[†] Nitin Chaudhary,[†] and Vibin Ramakrishnan^{*,†,‡}

[†]Department of Biosciences and Bioengineering and [‡]Department of Electronics and Electrical Engineering, Indian Institute of Technology Guwahati, Guwahati 781039, India

Supporting Information

ABSTRACT: The aggregation of β -amyloid peptides is a key event in the formative stages of Alzheimer's disease. Promoting folding and inhibiting aggregation was reported as an effective strategy in reducing $A\beta$ -elicited toxicity. This study experimentally investigates the influence of the external electric field (EF) and magnetic field (MF) of varying strengths on the in vitro fibrillogenesis of hydrophobic core sequence, $A\beta_{16-22}$, and its parent peptide, $A\beta_{1-42}$. Biophysical methods such as ThT fluorescence, static light scattering, circular dichroism, and infrared spectroscopy suggest that EF has a stabilizing effect on the secondary structure, initiating a conformational switch of $A\beta_{16-22}$ and $A\beta_{1-42}$ from β to non- β conformation. This observation was further corroborated by dynamic light scattering and transmission electron microscopic studies. To mimic in vivo conditions, we repeated ThT fluorescence assay with $A\beta_{1-42}$ in human cerebrospinal fluid to verify EF-mediated modulation. The self-seeding of $A\beta_{1-42}$ and cross-seeding with $A\beta_{1-40}$ to verify that the autocatalytic amplification of self-assembly as a result of secondary nucleation also yields comparable results in EF-exposed and unexposed samples. $A\beta$ -elicited toxicity of EF-treated samples in two neuroblastoma cell lines (SH-SY5Y and IMR-32) and human embryonic kidney cell line (HEK293) were found to be 15–38% less toxic than the EF-untreated ones under identical conditions. Experiments with fluorescent labeled $A\beta_{1-42}$ to correlate reduced cytotoxicity and cell internalization suggest a comparatively smaller uptake of the EF-treated peptides. Our results provide a scientific roadmap for future noninvasive, therapeutic solutions for the treatment of Alzheimer's disease.

KEYWORDS: Alzheimer's disease, electric field, magnetic field, diphenylalanine, amyloid β , perturbant

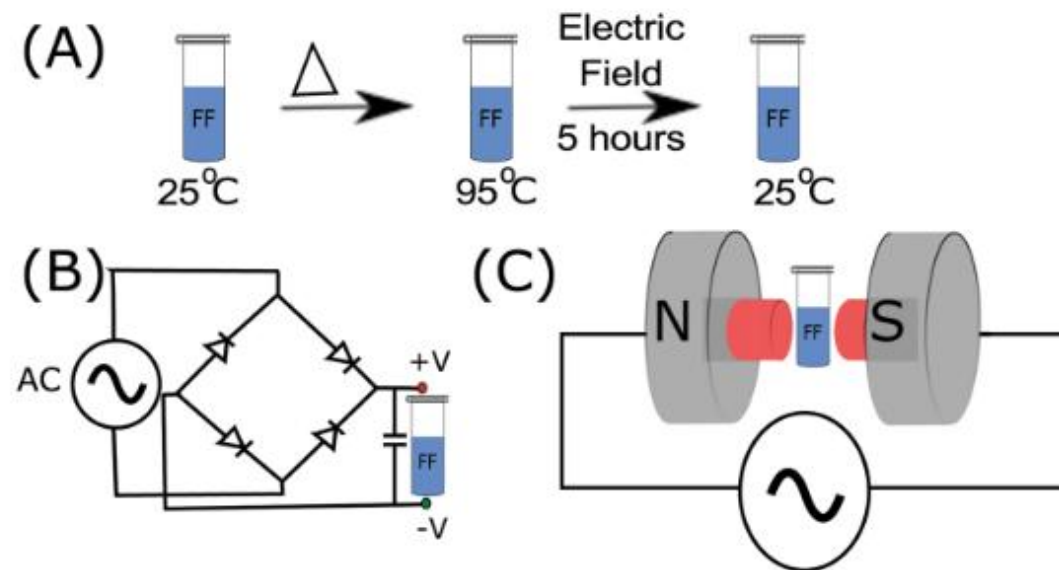
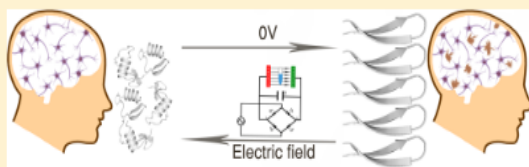
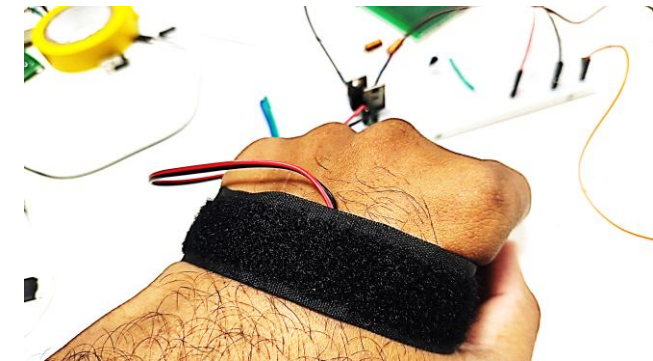
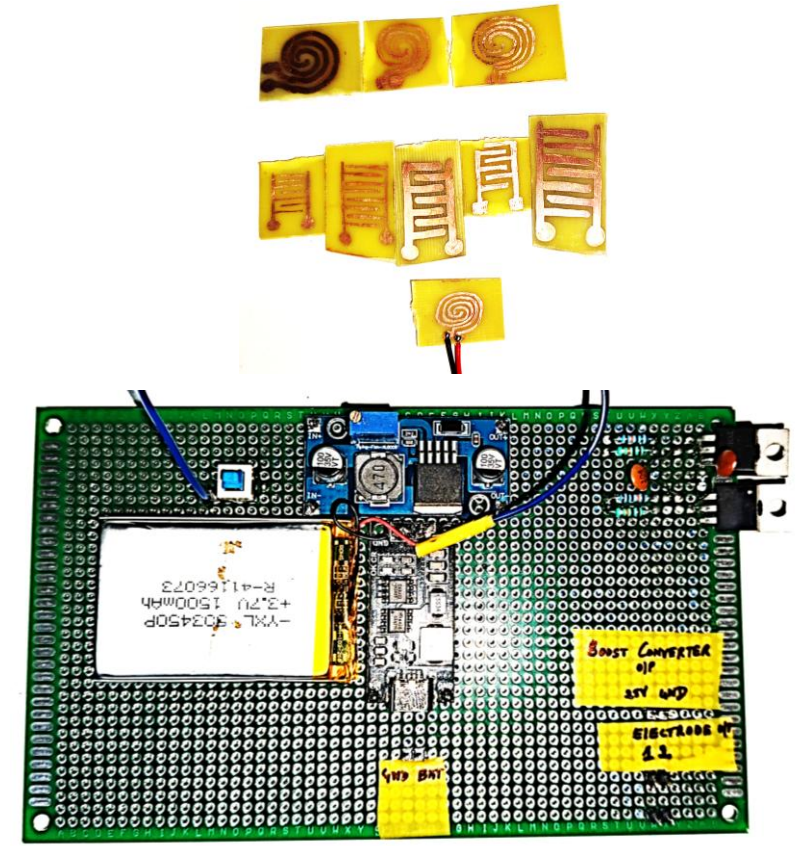


Figure 1. Schematic representation of experimental setup. (A) Dissolved peptide was allowed to cool down from 95 °C to room temperature while incubating in the presence of electric field. (B) Electric field set up and (C) magnetic field setup.

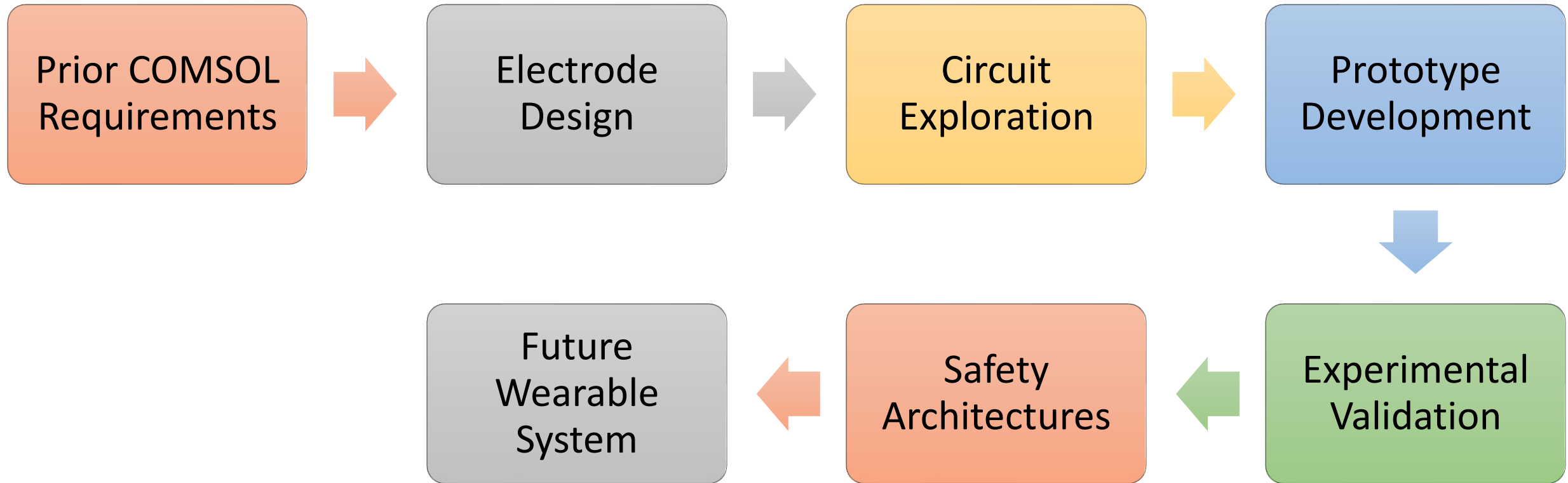
- Electric fields can inhibit misfolded protein aggregation
- In-vitro: **150 V/cm slows β -amyloid aggregation**
- Neuromodulation aligns molecular dipoles
- External fields interact with neuron-generated potentials
- Potential to delay disease progression
- Attempts to modulate biophysical modulation of aggregation pathways

Research Objectives

- Design of Stimulation Electrode
- Develop Stimulation Circuitry
- Build Prototype or Proof-of-Concept
- Validate Experimentally



Overall Methodology



Prior Work and Design Requirements

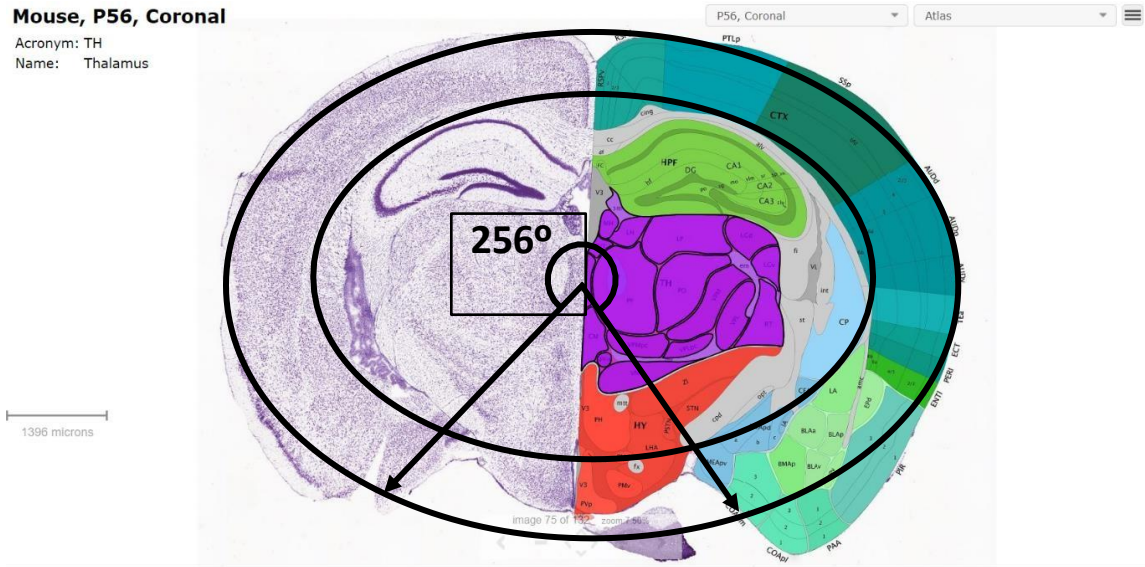


Figure 4. Model of mouse brain depicting the location of thalamus and angle of coverage required for effective stimulation

Degrees of stimulation effective to stimulate the brain effectively considered in previous phase of work for COMSOL Simulations (Adapted from [Allen Brain Institute](#))

Prior COMSOL Studies:

- 10 electrode concept
- Width : Gap = 1
- 256° Angular Coverage
- Target Field of 150 V/cm
- 25 V Capable of generating required field

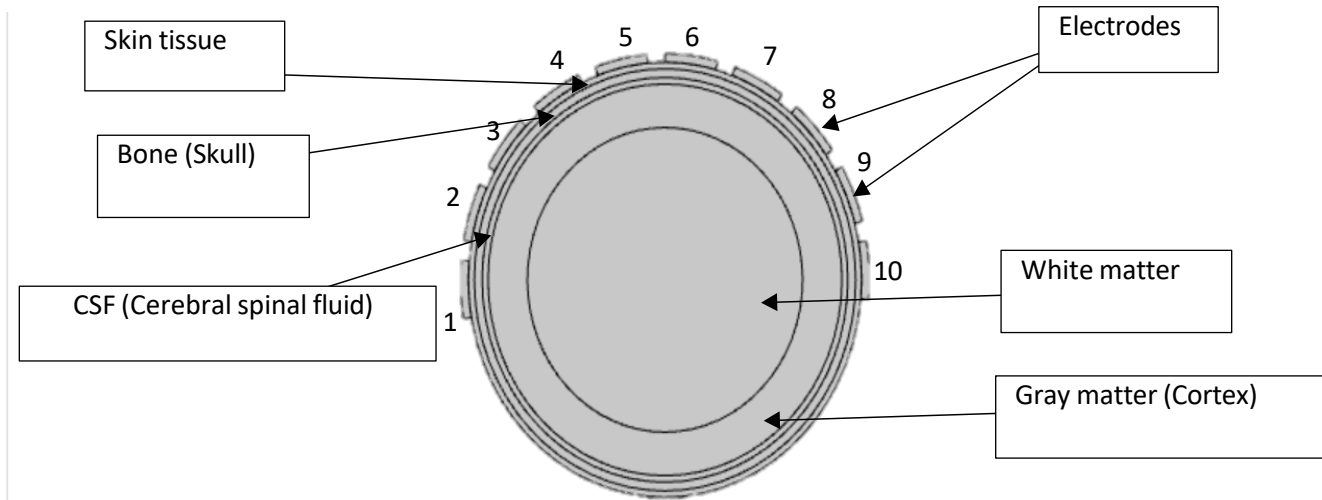


Figure 5. Parts of Human Head considered for COMSOL Simulations

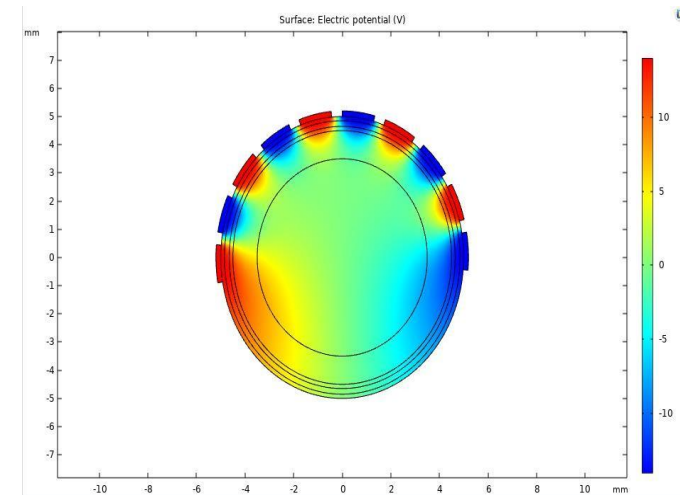


Figure 6. Electric Field Distribution for 10 Electrodes

Note. Figure 4 & Figure 5 are adapted from Development of Device for Non-Invasive Treatment of Alzheimer's Disease (AD) Using Electric Field (EF), by Pooja, 2024, Indian Institute of Technology Guwahati.

Electrode Design Evolution

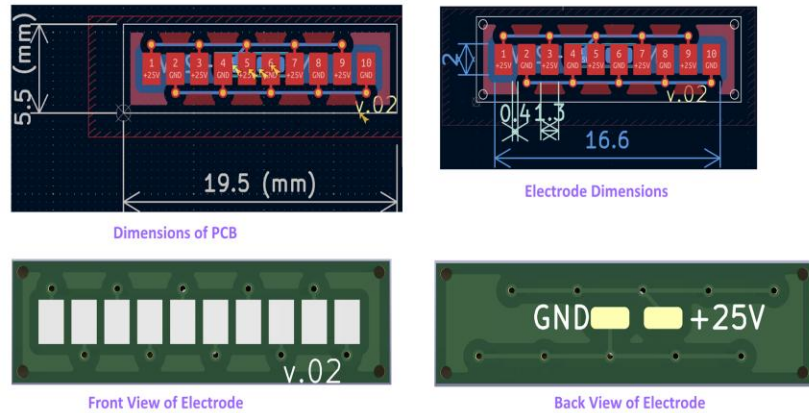


Figure 7. Version 02 of electrode design

Effective stimulation of 1.66 cm in length and 0.2 cm in breadth

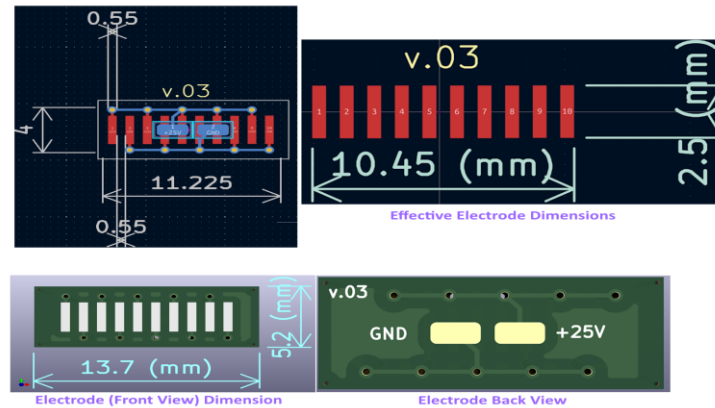


Figure 8. Version 03 of electrode design

Effective stimulation of 1.45 cm in length and 0.25 cm in breadth

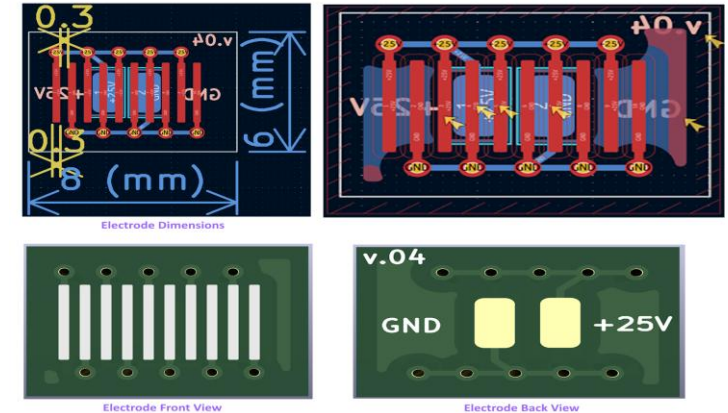


Figure 9. Version 04 of electrode design

Dimensions of 0.8 cm in length and 0.6 cm in breadth

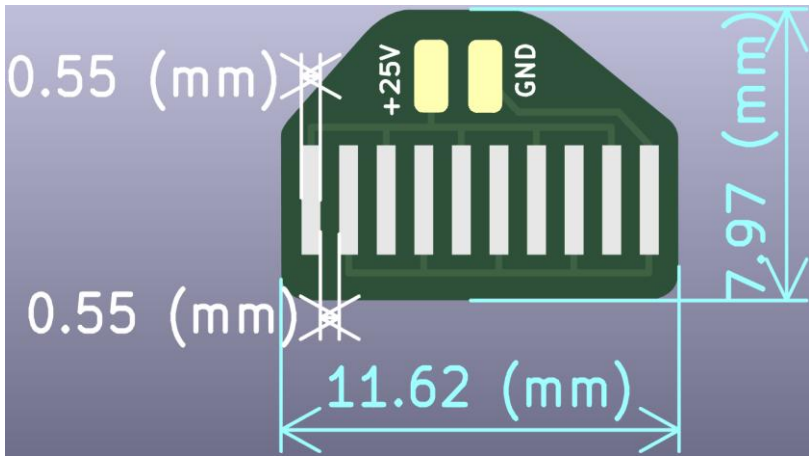


Figure 10. Version 05 of electrode design

Dimensions of 1.16 cm in length and 0.79 cm in breadth

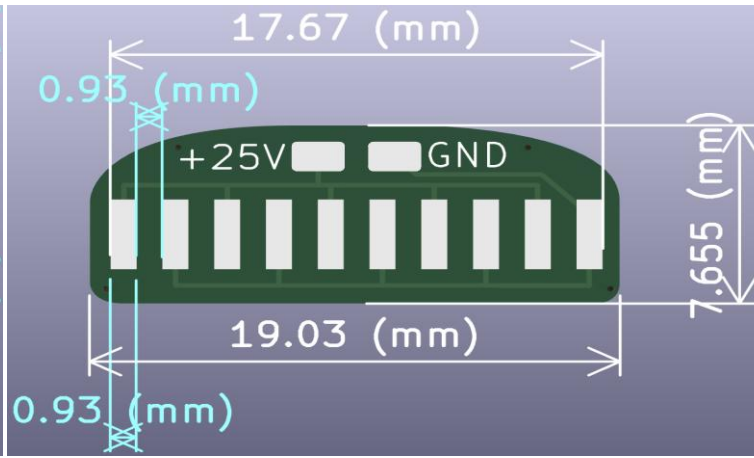


Figure 11. Version 06 of electrode design

Dimensions of 1.45 cm in length and 0.25 cm in breadth

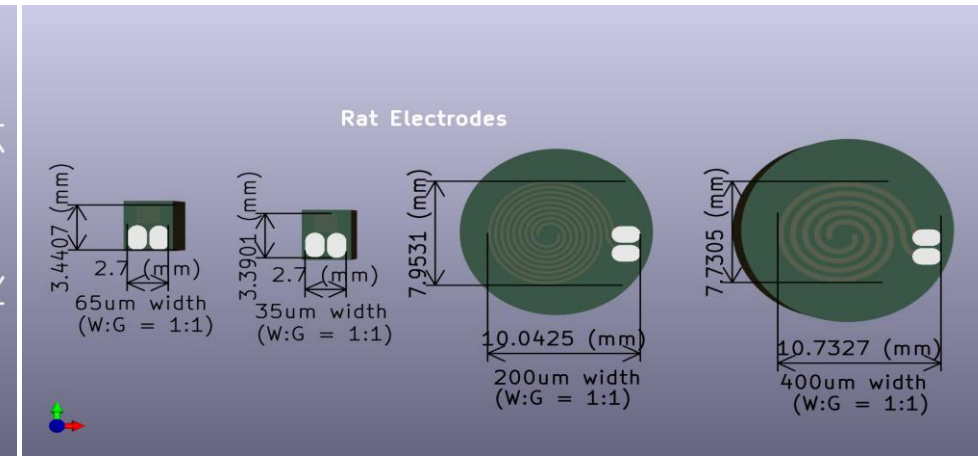
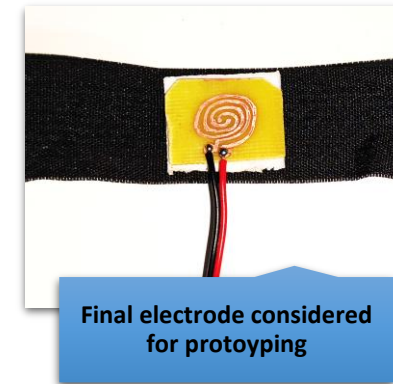


Figure 12. Version 07 of electrode design

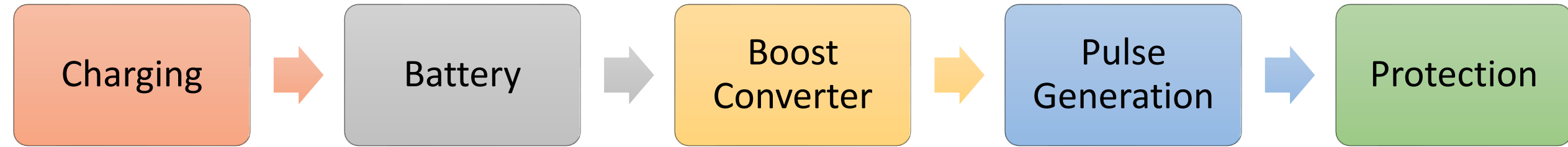
Considered for prototyping

Electrode Fabrication for Prototyping

- Although multiple electrode geometries were developed, the spiral configuration was selected for proof-of-concept implementation due to its ability to preserve the simulation derived width-to-gap relationship while offering a practical balance between available simulation area, mechanical robustness and manufacturability.
- Due to the temporary unavailability of PCB fabrication facilities during the course of the study, an alternative fabrication approach was adopted to obtain a proof-of-concept electrode for preliminary experimental evaluation.



Initial System Architecture



System Requirements :

- Compact and light weight architecture
- Low component count
- Battery powered operation
- Ease of troubleshooting and experimental validation
- Reliable generation of biphasic stimulation

Power Conversion Architecture

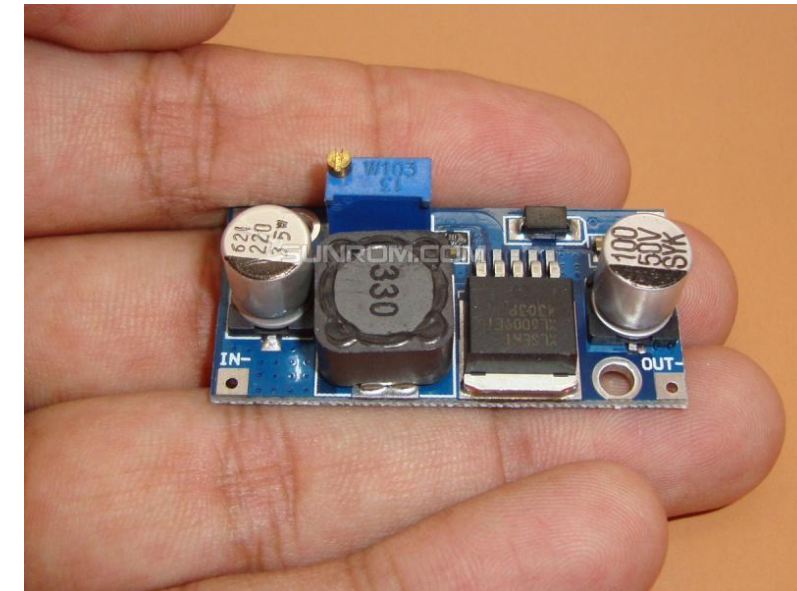
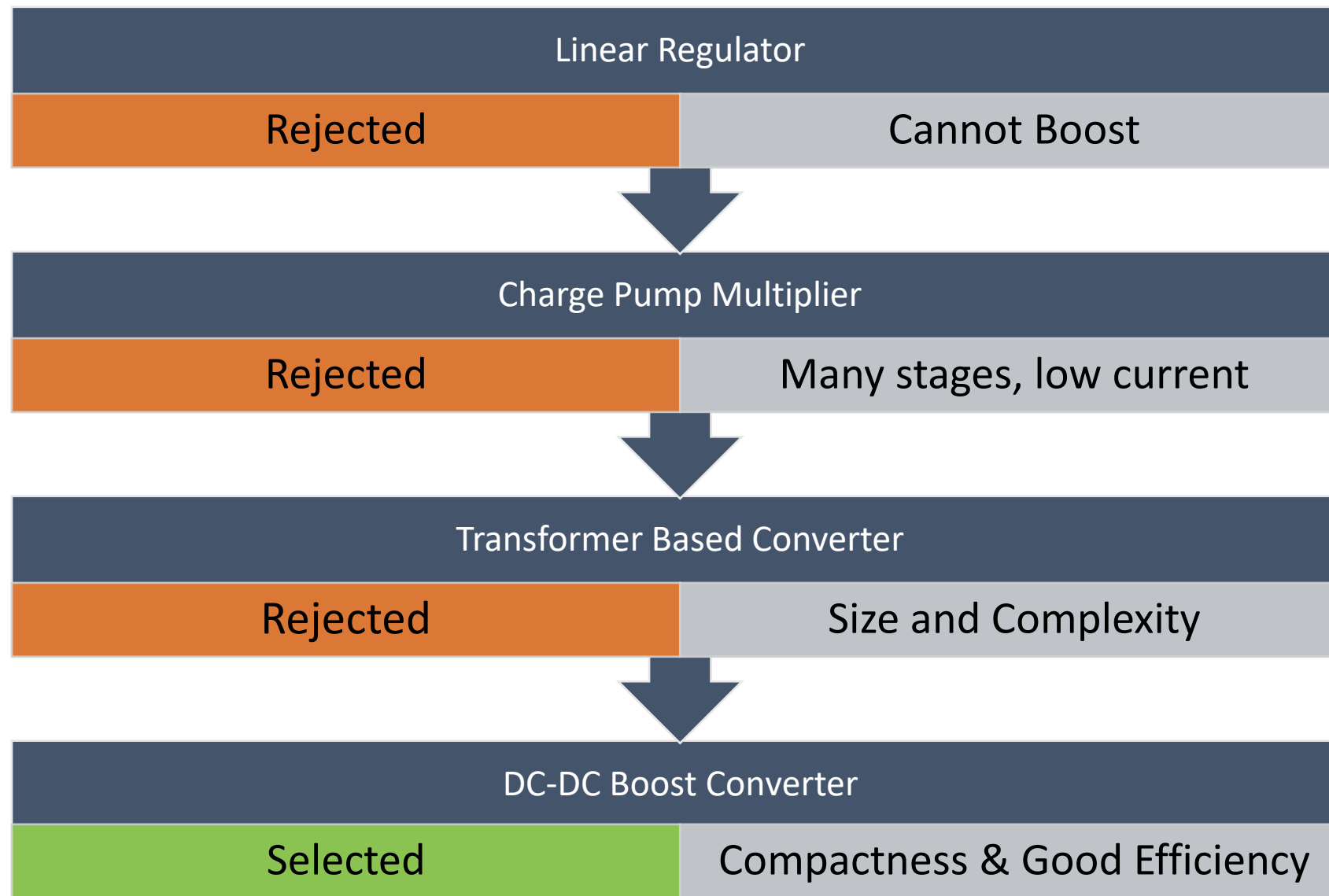


Figure 13. XL6009 DC-DC Boost converter module currently used due to lack of availability of MT3068 at the time of prototyping

Small size : 1.29inx0.70in

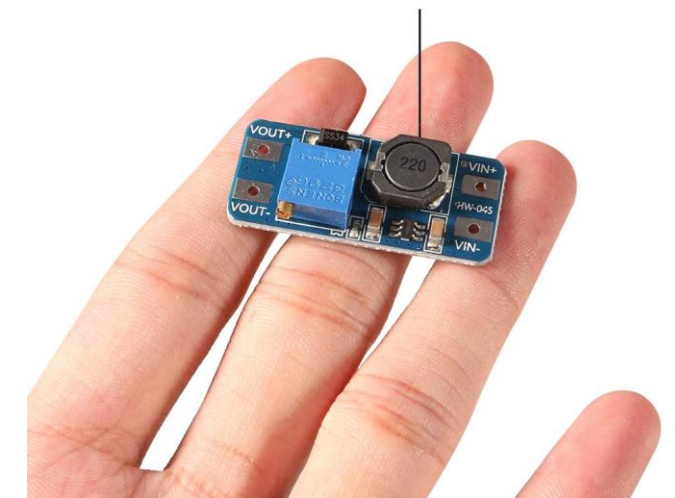


Figure 14. Suggest MT3608 for future considerations

Circuit Design Evolution

Circuit	Implemented
BC547 Astable	✓
2N2222A Astable	✓
LM358 Oscillator	✓
Bridge Tied Load	✓
Astable H-Bridge	✓
Self Oscillating H-Bridge	✓
IRF540N Astable	✓

BC547 to 2N2222A Evolution

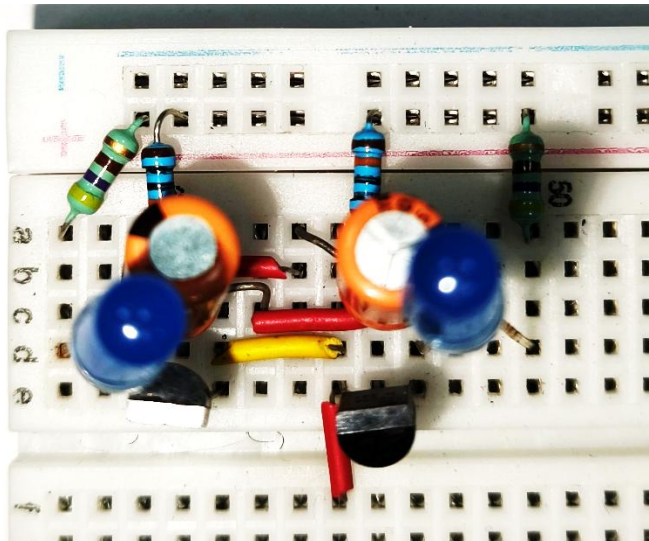


Figure 15. BC547 breadboard implementation for Preliminary testing.

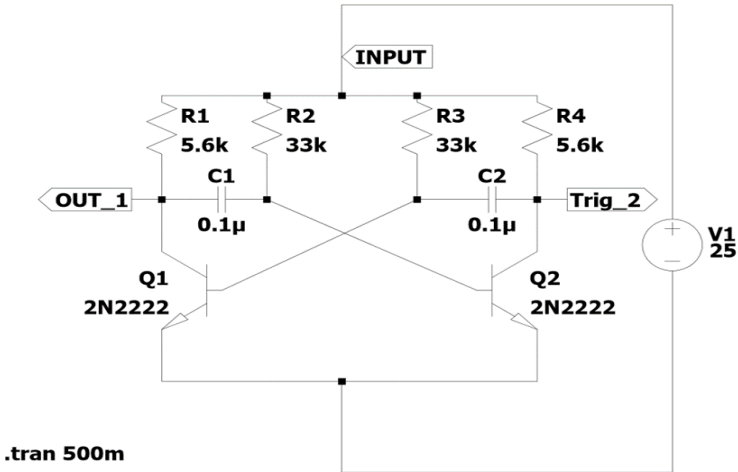
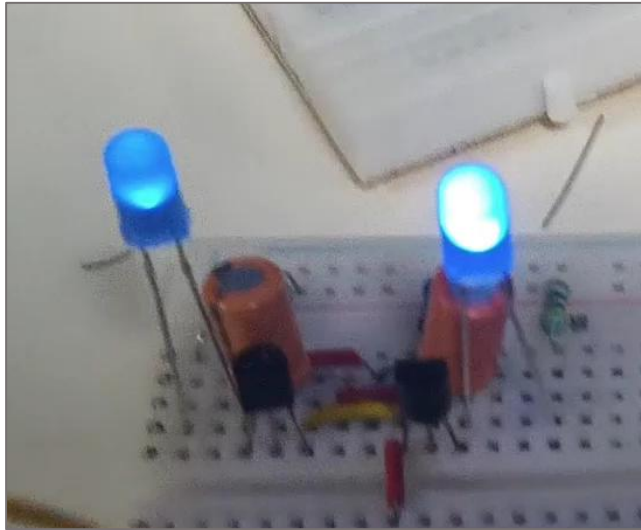


Figure 16. LTSpice schematic of the 2N2222A astable multivibrator.

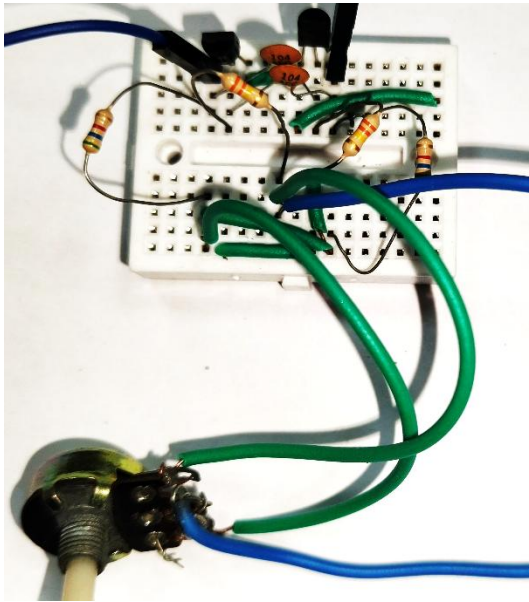
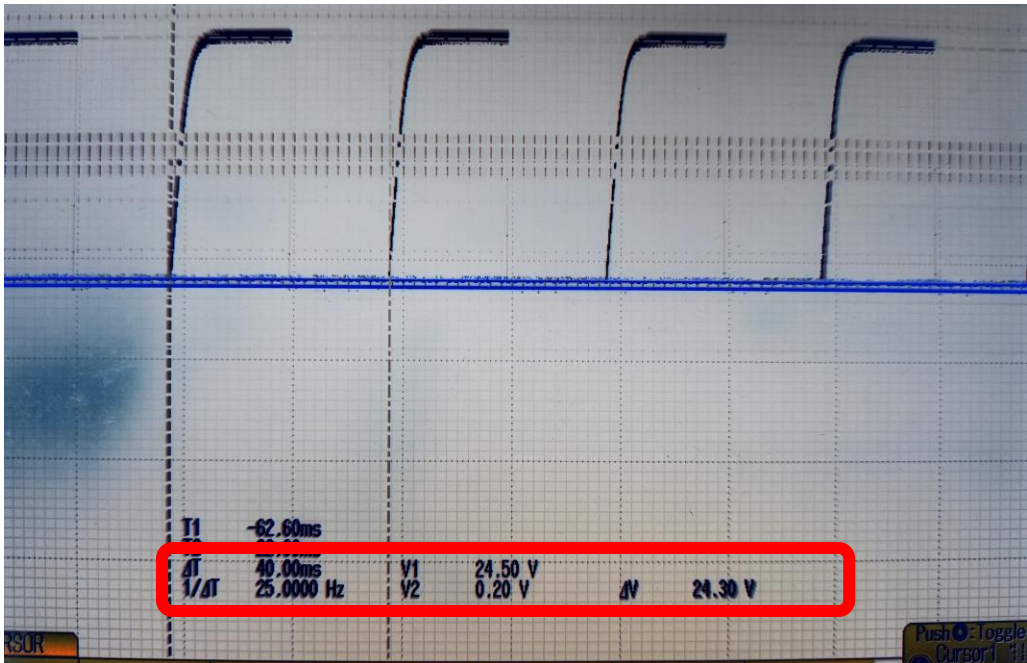


Figure 17. Breadboard implementation of the 2N2222A astable multivibrator.

Collector Resistors	Base Resistors	Observed Output Voltage
1 kΩ	10 kΩ	5.6 V
10 kΩ	100 kΩ	12.2 V
47 kΩ	470 kΩ	16.2 V
470 kΩ	680 kΩ	6.2 V
51 kΩ	470 kΩ	16.8 V
51 kΩ	680 kΩ	16.6 V
5.6 kΩ	33 kΩ	≈ 25 V

Figure 18. Oscilloscope waveform of 2N2222A astable Multivibrator.

LM358 based Astable Multivibrator

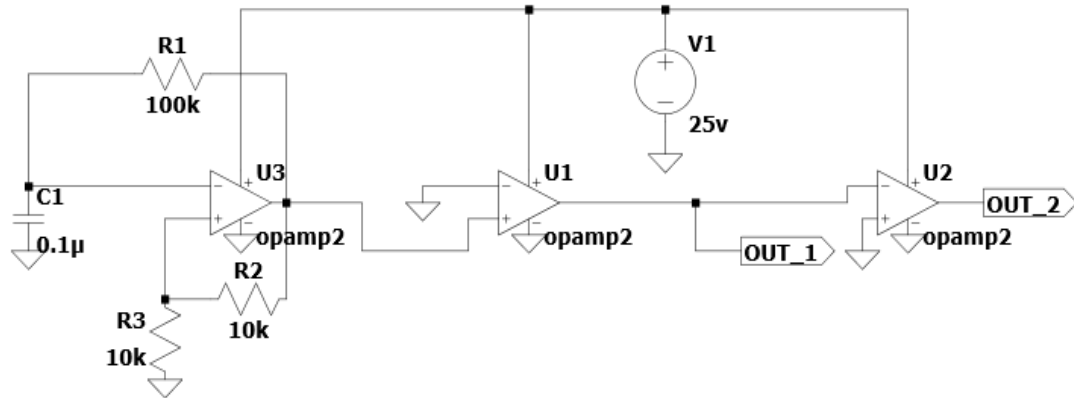


Figure 19. LTSpice schematic of the comparator based BTL architecture.

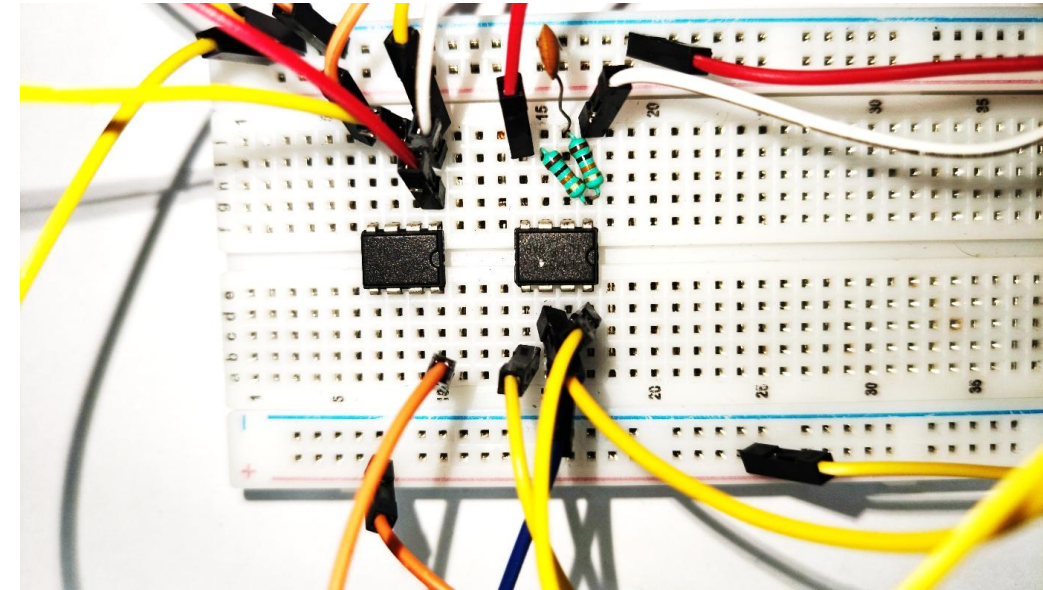


Figure 20. Breadboard implementation of the BTL architecture.

- **Oscillations were generated successfully.**
- **Biphasic outputs were obtained using the BTL configuration.**
- **Waveform distortion increased during practical implementation.**
- **Increased sensitivity to breadboard parasitic and capacitance and wiring effects.**
- **Limited output swing near supply rails.**
- **Additional active components increased circuit complexity**

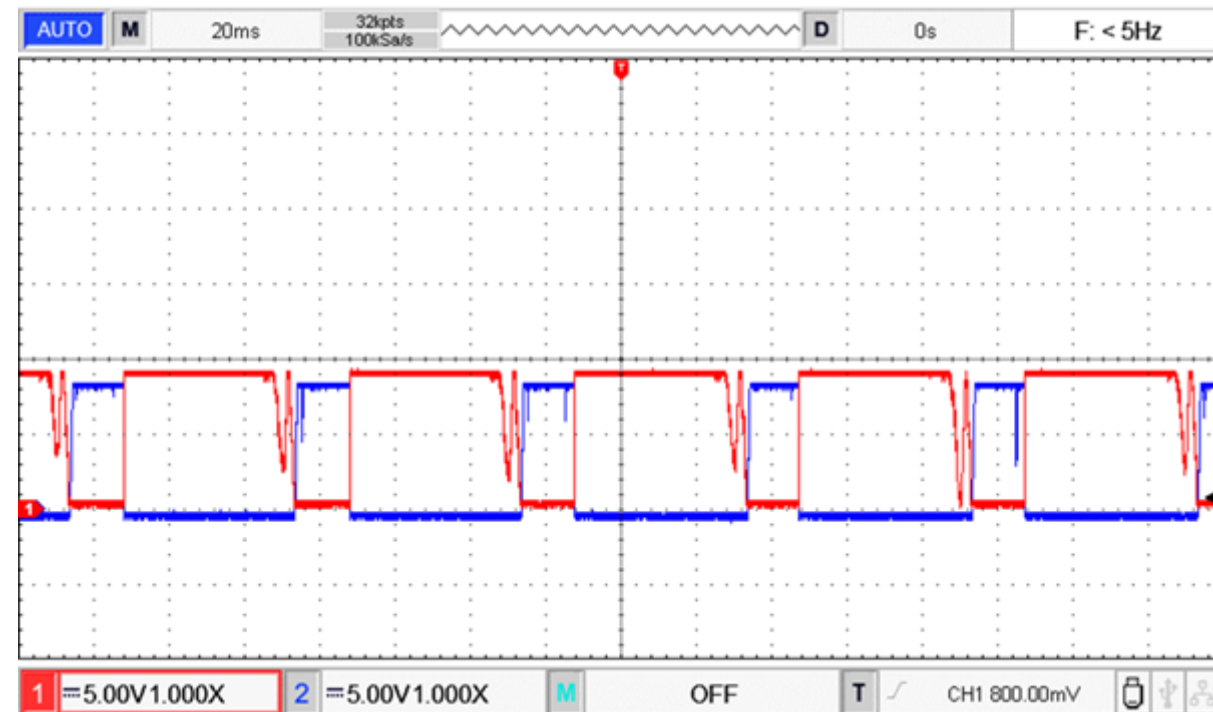
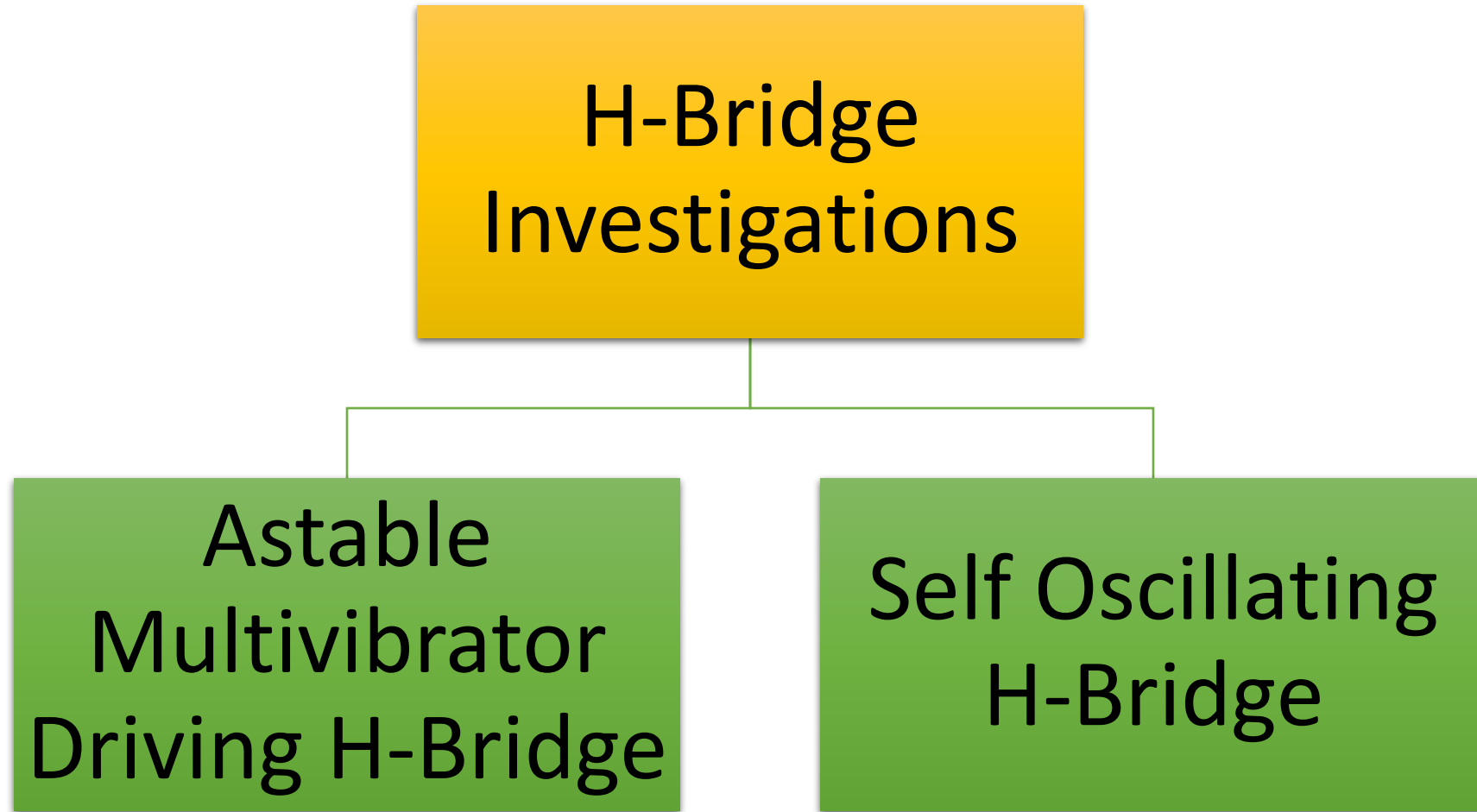


Figure 21. Oscilloscope waveform of oscillator output obtained from the implemented BTL architecture.

H-Bridge Investigations



Astable Multivibrator Driving a MOSFET H-Bridge

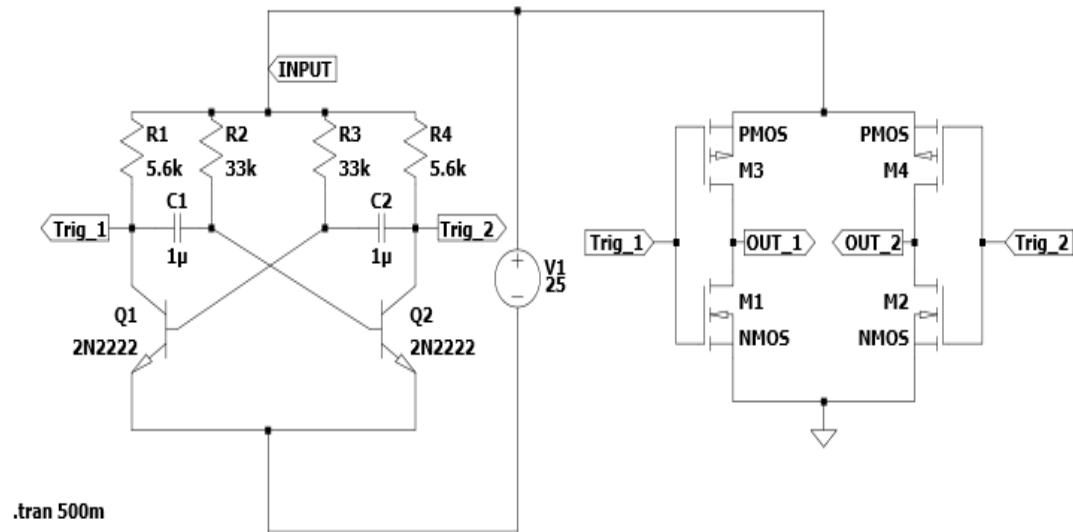


Figure 22. LTSpice Schematic of the transistor-driven MOSFET H-Bridge.

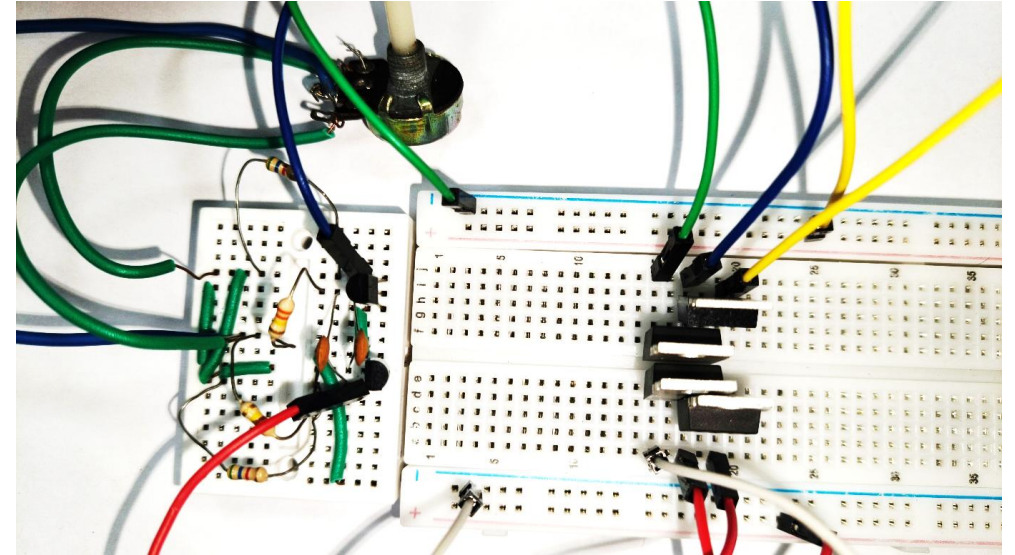


Figure 23. Breadboard implementation of the power MOSFET H-Bridge architecture.

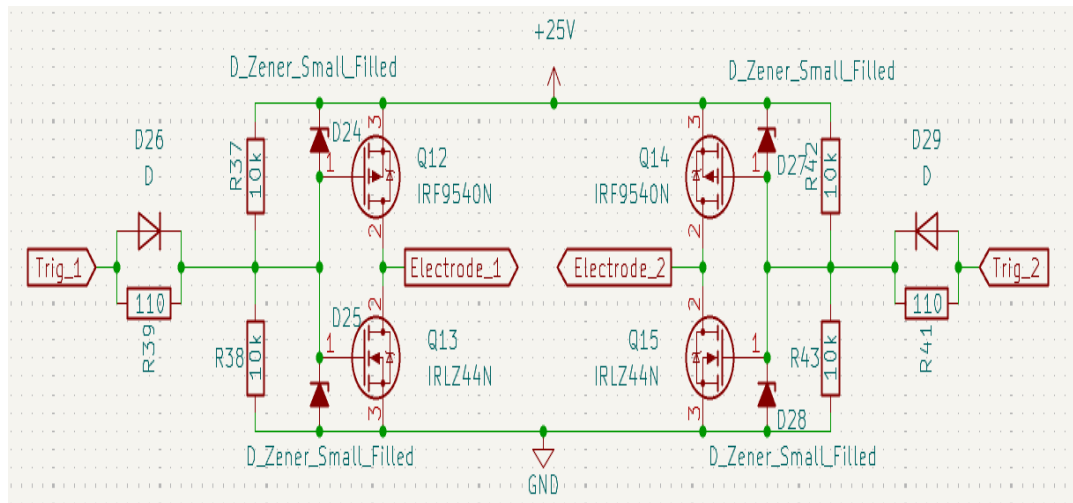


Figure 24. Improved MOSFET H-Bridge architecture.

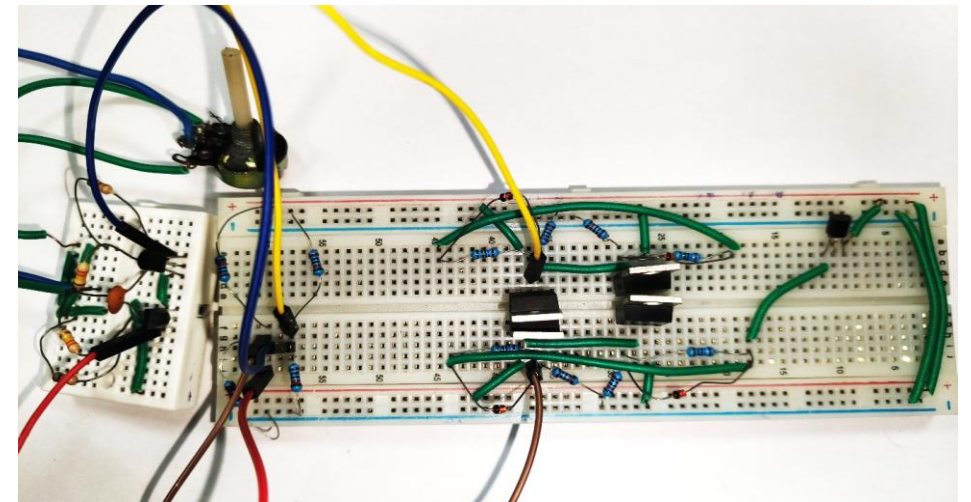


Figure 25. Breadboard implementation of the improved MOSFET H-Bridge architecture.

Astable Multivibrator Driving a MOSFET H-Bridge

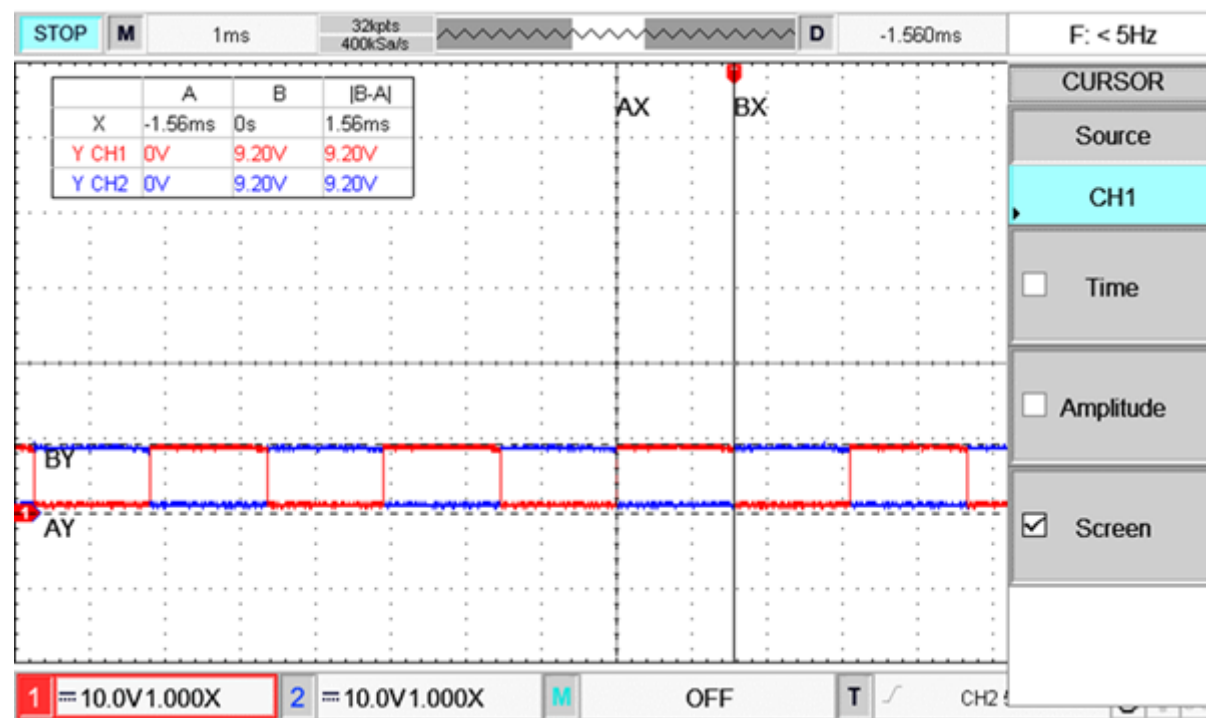
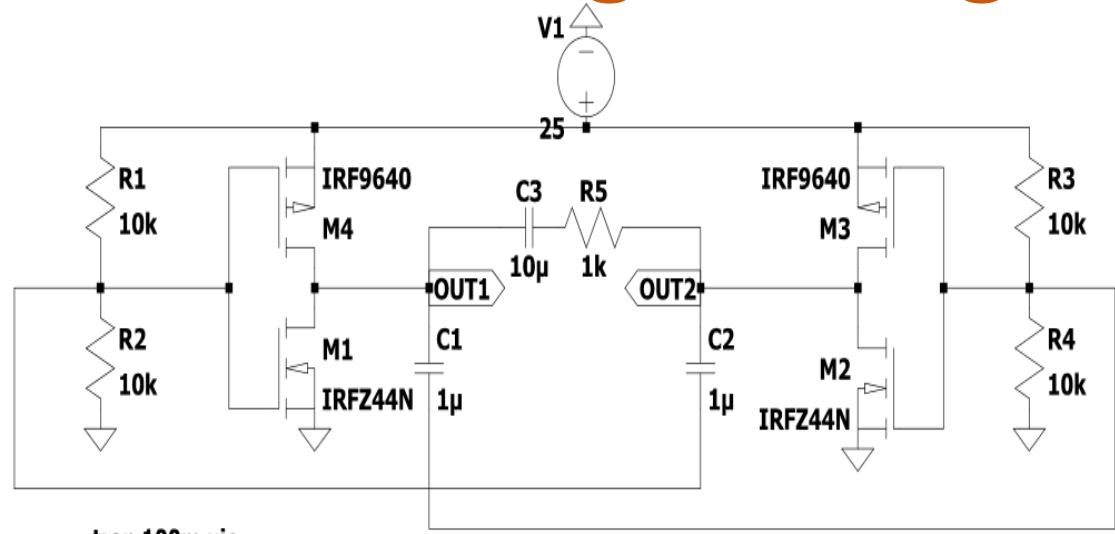


Figure 26. Oscilloscope waveforms at the output stage of Astable Multivibrator driven H-Bridge in the improved MOSFET H-Bridge architecture.

Self-Oscillating H-Bridge



.tran 100m uic

Figure 27. LTSpice schematic of Self-Oscillating MOSFET H-Bridge.

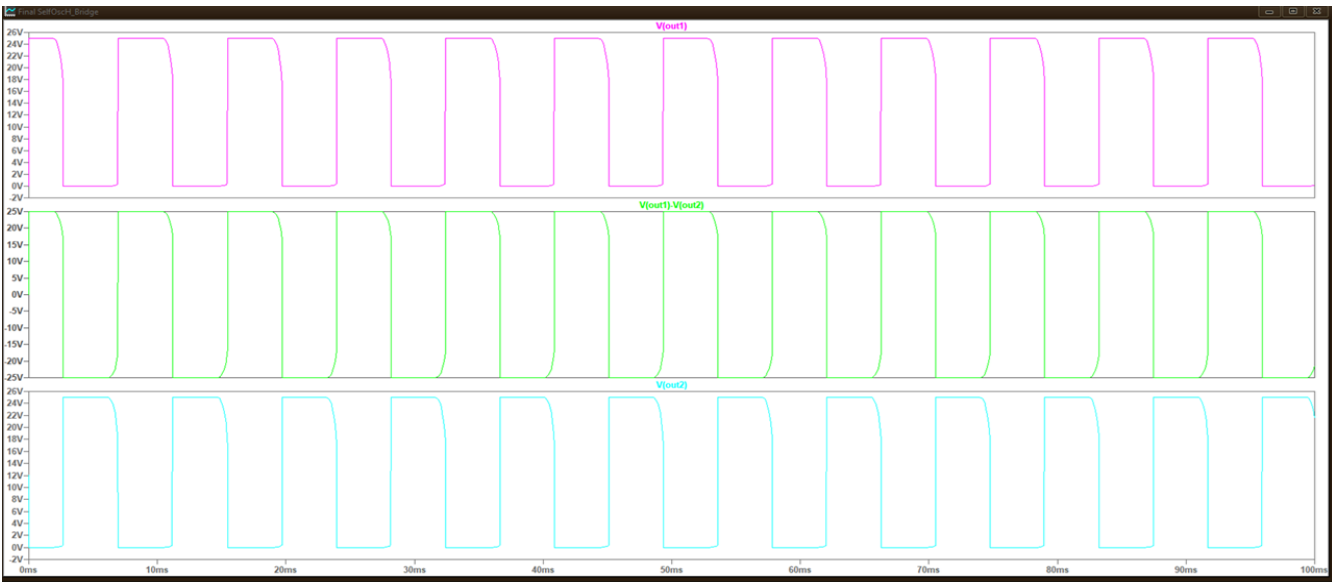


Figure 28. LTSpice simulated waveforms of output of Self-Oscillating MOSFET H-Bridge.

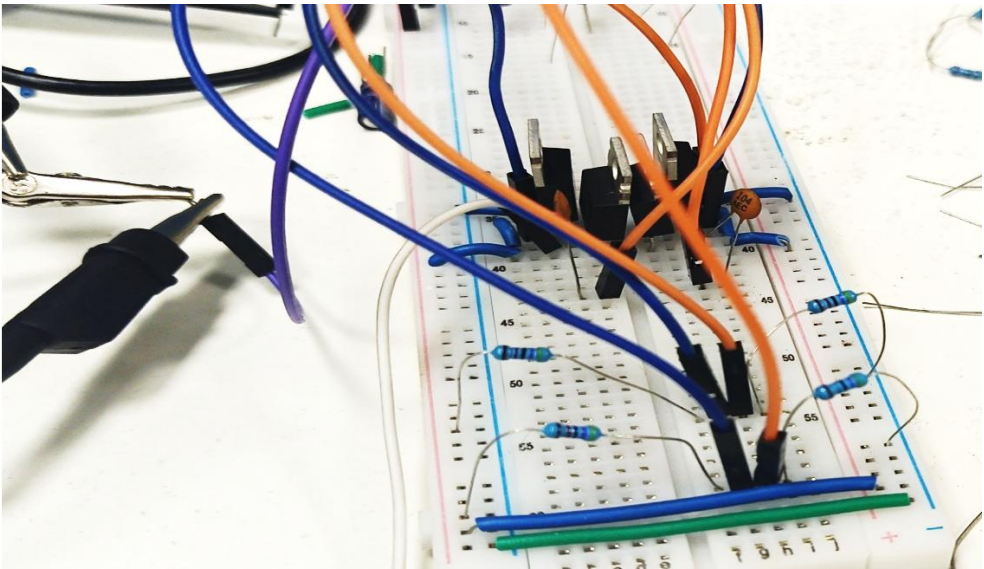


Figure 28. Breadboard implementation of Self-Oscillating MOSFET H-Bridge.

Gate Resistors	Capacitor	Output Voltage
100 k Ω	0.1 μ F	8.4 V
100 k Ω	1 μ F	10.4 V
10 k Ω	0.1 μ F	9.2 V
10 k Ω	1 μ F	8.2 V
10 k Ω	2.2 μ F	8.2 V
10 k Ω	3.3 μ F	~10 V
10 k Ω	4.7 μ F	~11.2 V
10 k Ω	10 μ F	9.6 V
20 k Ω	100 μ F	13.6 V
100 k Ω	220 μ F	~18.8 V

Self-Oscillating H-Bridge

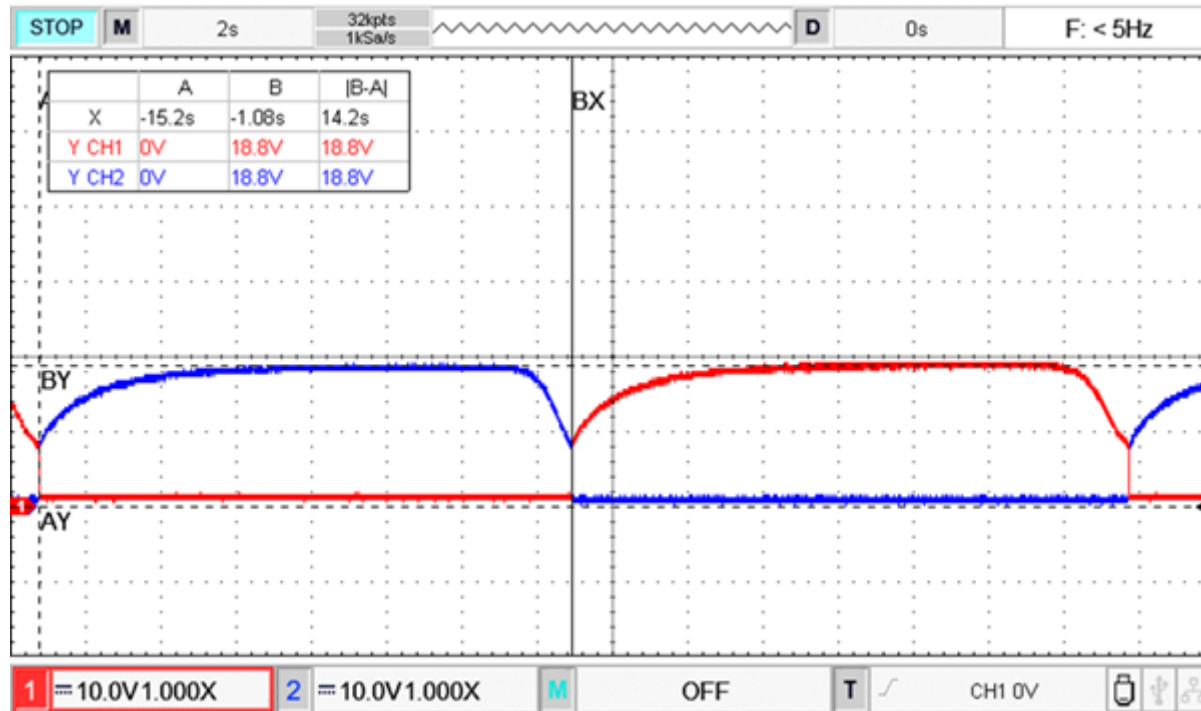


Figure 29. Oscilloscope waveforms at output stage of implemented Self-Oscillating MOSFET H-Bridge.

Despite multiple design iterations including gate delay networks, gate driver stages, resistor-capacitor optimization, mid point gate biasing, pull-up/ pull-down networks and Zener protection circuitry, stable operation could not be consistently achieved. Therefore, further H-bridge development was discontinued and attention was directed toward simpler architectures that demonstrated reliable biphasic waveform generation.

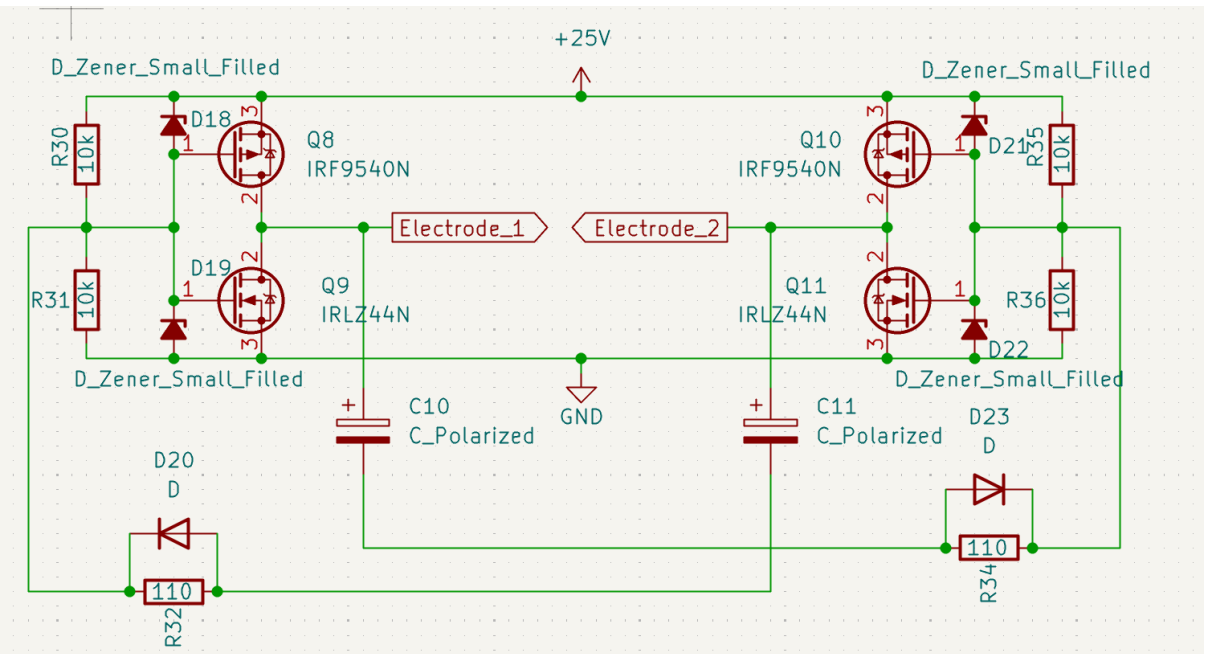


Figure 30. Improved version of Self-Oscillating MOSFET H-Bridge architecture.

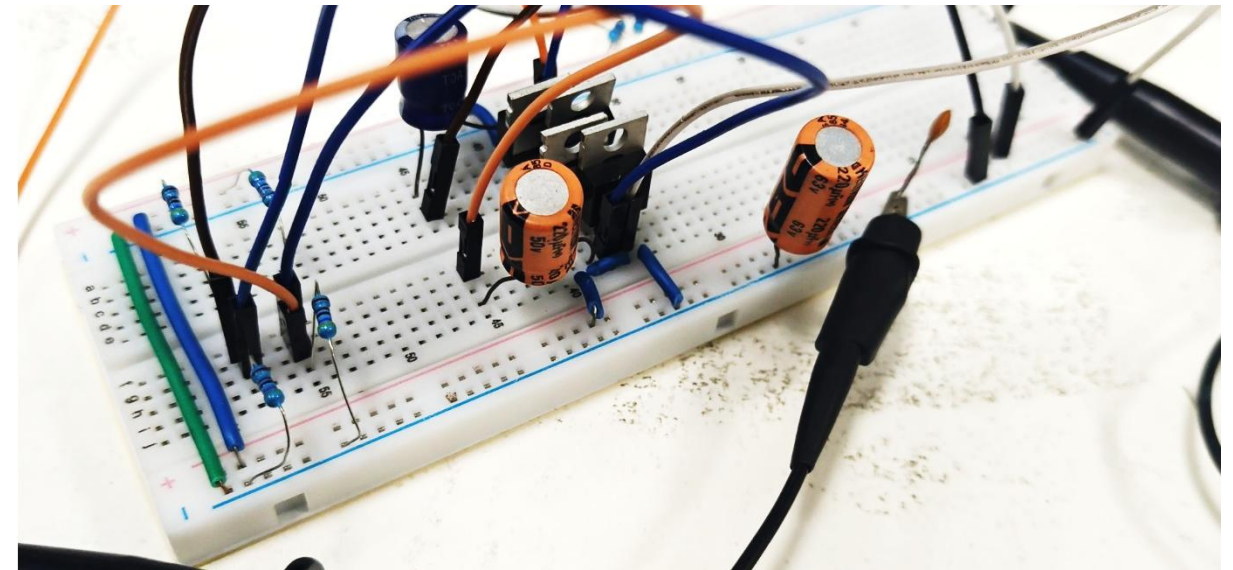


Figure 31. Breadboard implementation of improved Self-Oscillating MOSFET H-Bridge architecture.

Selection Rationale for Final Implementation

Parameter	BJT (BC547 / 2N2222A)	MOSFET (IRF540N)	Reason for Final Selection
Control Mechanism	Current Controlled	Voltage Controlled	MOSFET preferred
Input Impedance	Low to Moderate	Very High	MOSFET preferred
Drive Requirement	Continuous base current required	Only gate charging current required	MOSFET preferred
Static Power Consumption	Higher	Lower	MOSFET preferred
Switching Speed	Moderate	Higher	MOSFET preferred
Scalability for Future Current Levels	Limited	Better suited	MOSFET preferred
Compatibility with Protection Circuits	Additional drive circuitry often needed	Easier integration	MOSFET preferred
Operation at 25 V	Achieved with 2N2222A	Achieved with IRF540N	Both acceptable
Experimental Output	Successful oscillation	Successful oscillation with desired biphasic output	MOSFET selected

Final Implementation : IRF540N-based Astable Multivibrator

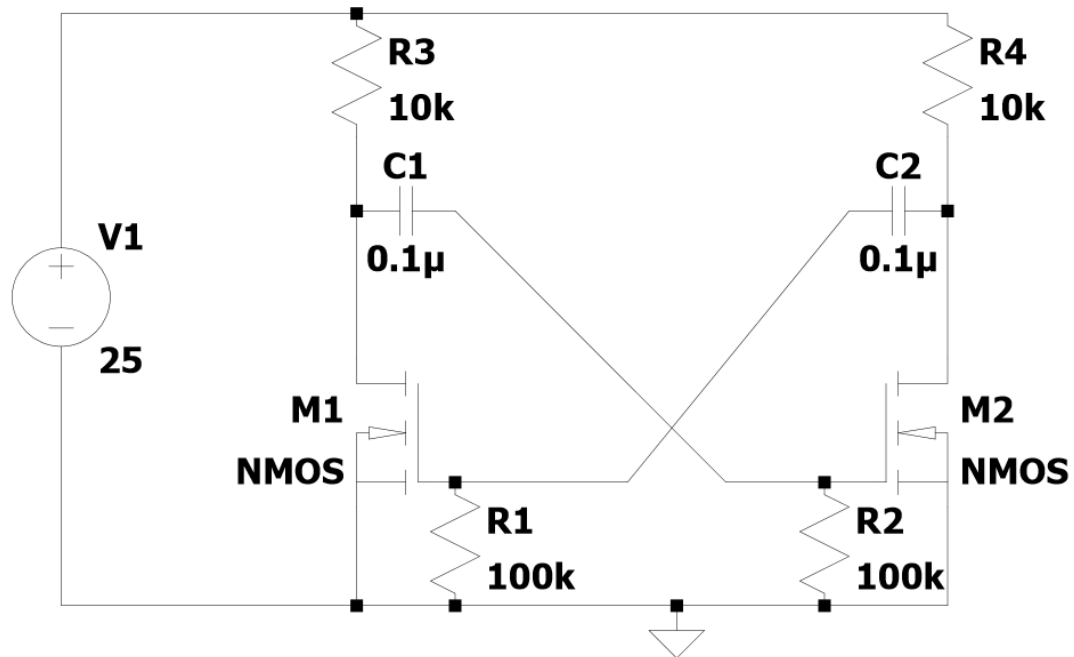


Figure 34. Self-Oscillating MOSFET H-Bridge troubleshooting breadboard implementation.

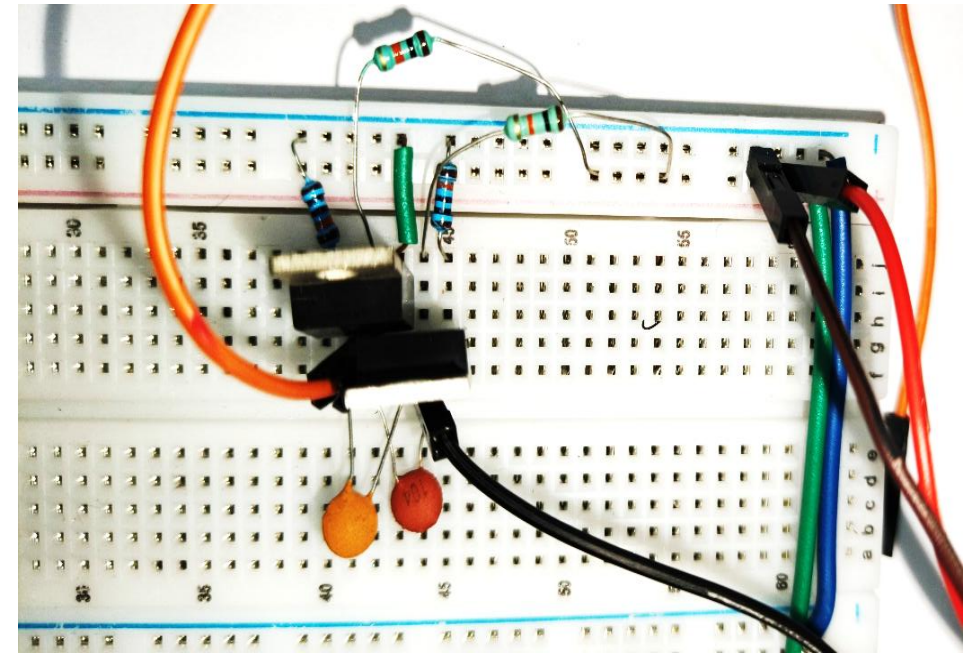


Figure 35. Breadboard implementation of IRF540N astable multivibrator

PCB Design for Proof-Of-Concept Implementation

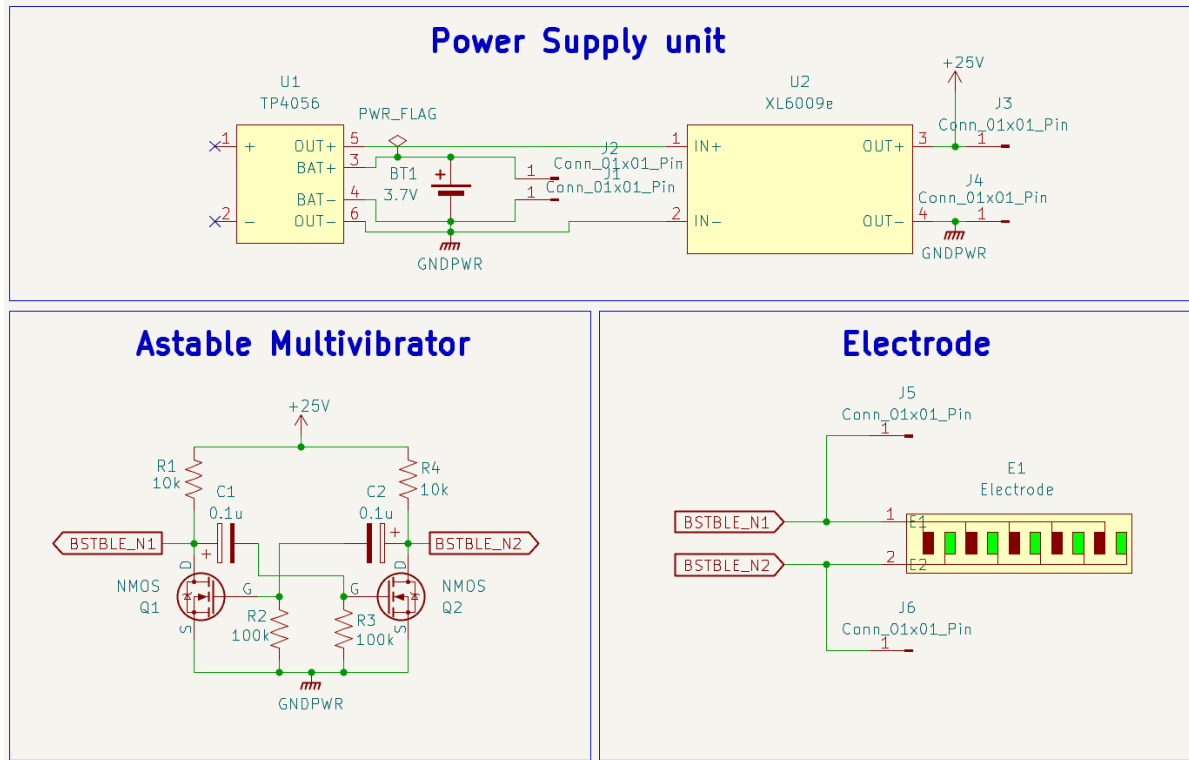


Figure 36. PCB schematic of Final Prototype Implementation

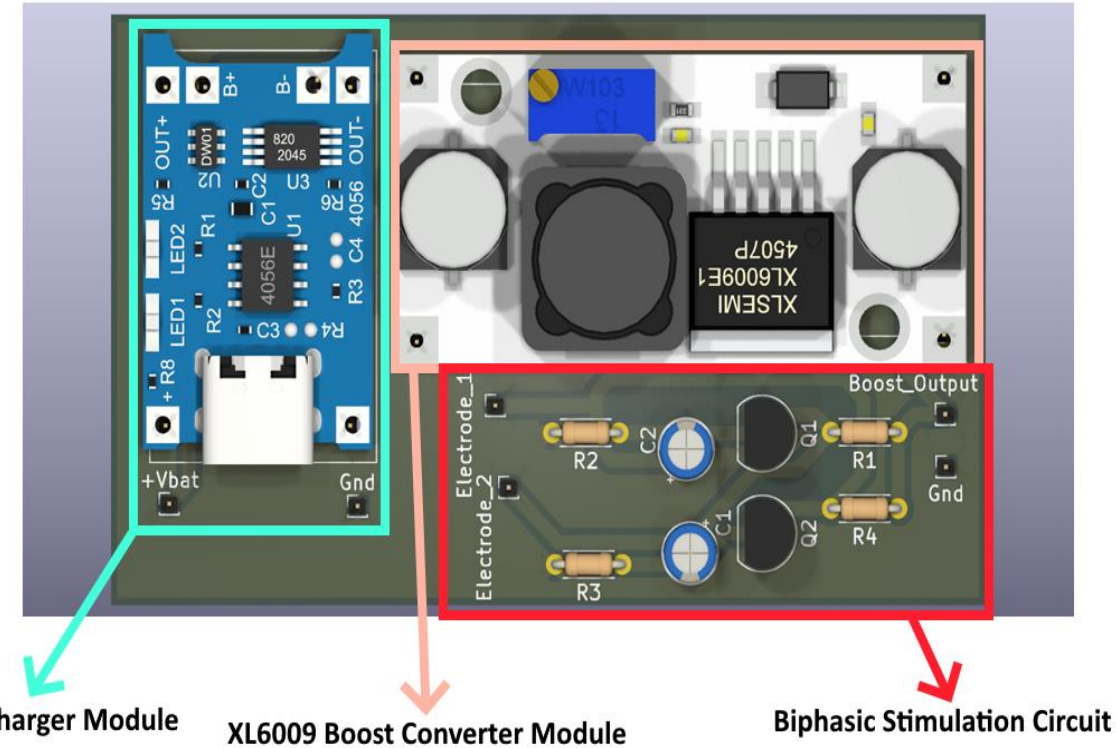


Figure 37. PCB designed for proof of concept Implementation

Proof-Of-Concept

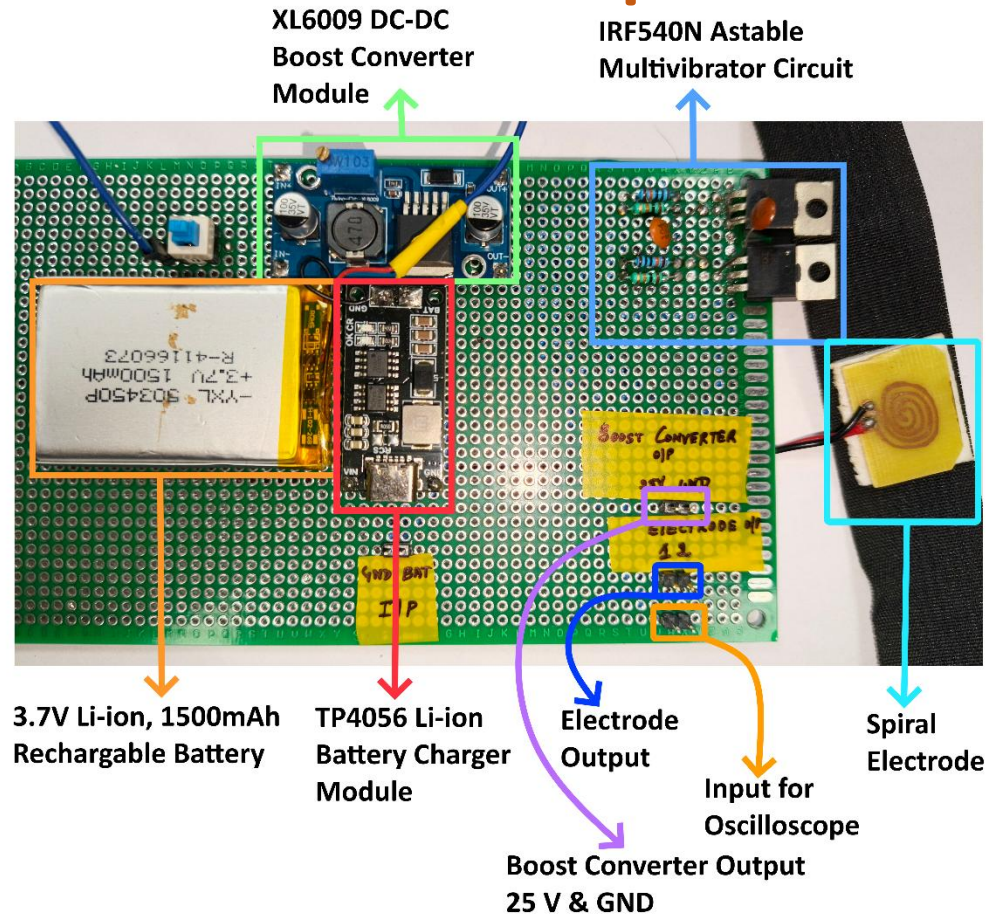


Figure 38. Prefboard or Zero PCB implementation of IRF540N astable multivibrator

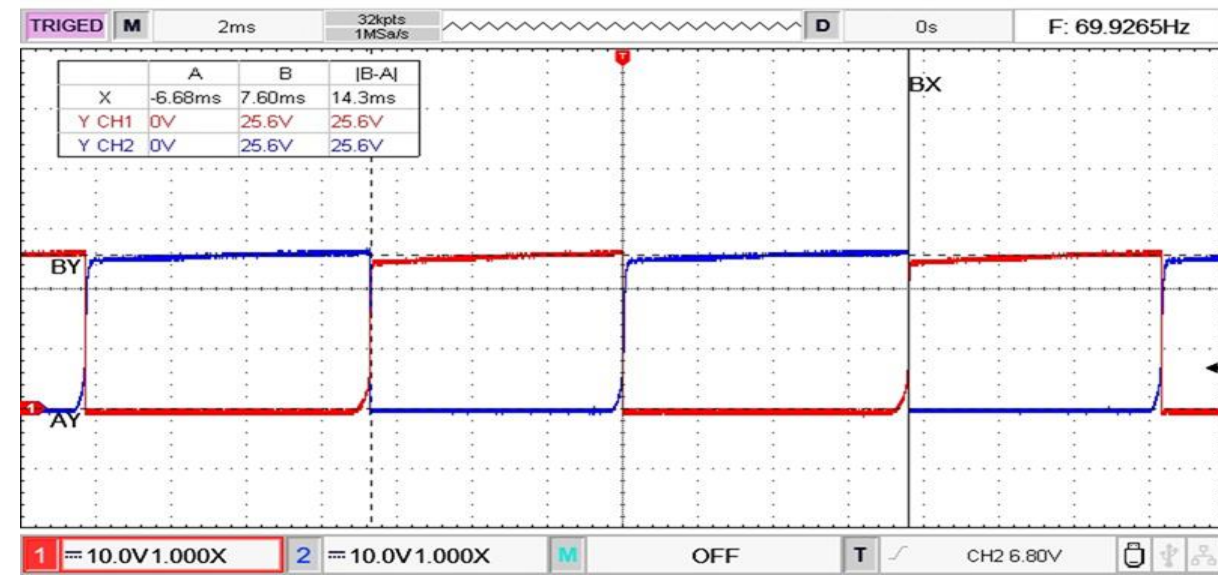


Figure 39. Oscilloscope waveforms at the electrode terminals of Final IRF540N Astable Multivibrator, showing amplitude of 25.6 V on each electrode terminal and frequency of 66 Hz.

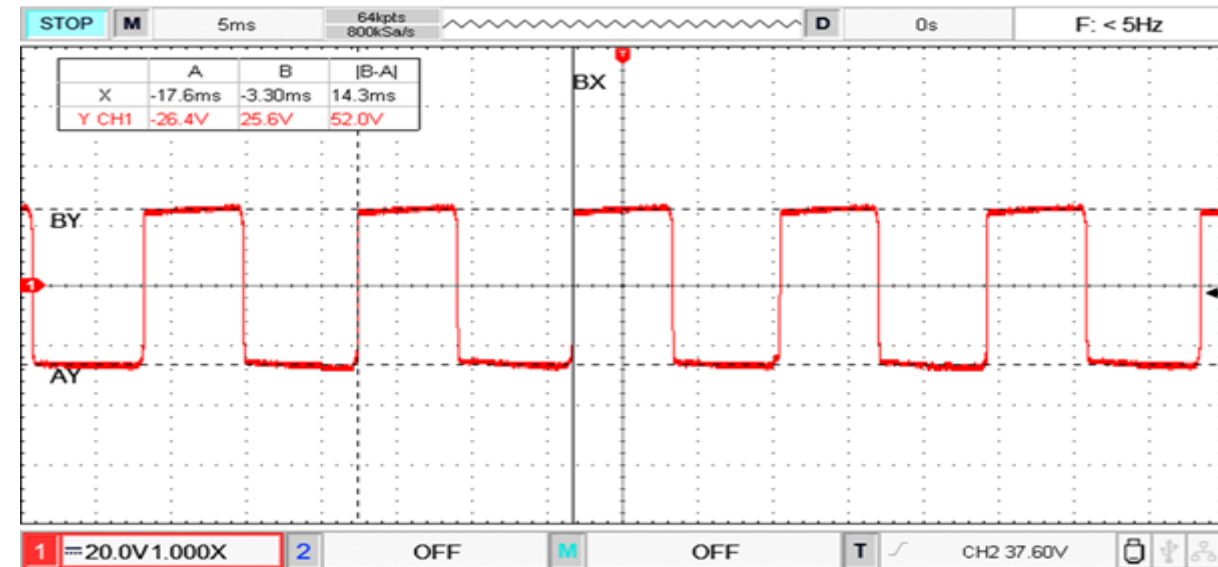
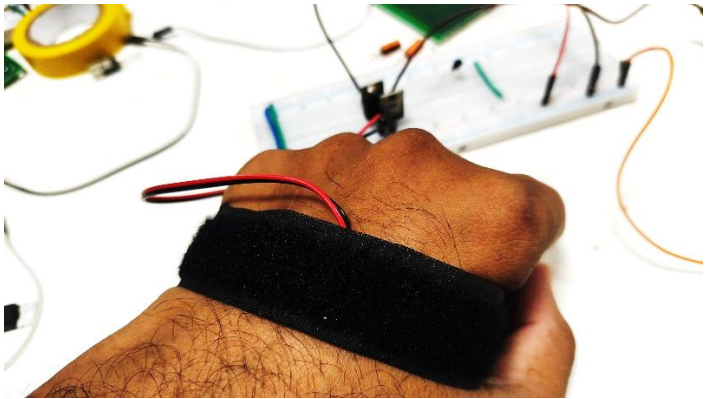


Figure 40. Oscilloscope waveform of differential voltage appearing across the load from the implementation of Final IRF540N Astable Multivibrator. The obtained output is ± 26 V, i.e. 52 V peak-to-peak at 70 Hz, fulfilling the objectives of the project.

Electrode Loading Evaluation



Condition	Peak-to-Peak Voltage (V)	Period (ms)
Electrode unloaded	52.8	11.5
Electrode placed on Skin	42.4	13.5

Loading was expected because biological tissue introduces conductive loading and parasitic capacitance.

The key observation is that Biphasic operation was preserved under loading conditions.

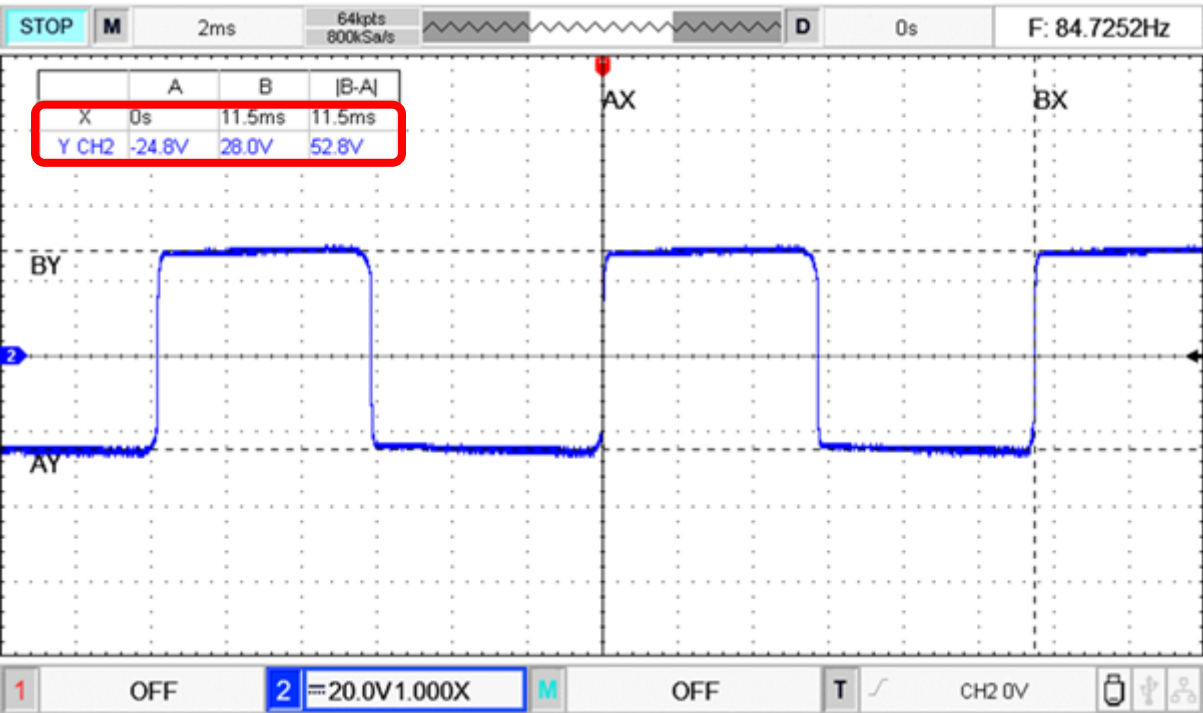


Figure 38. Unloaded Waveform (Electrodes without skin contact)

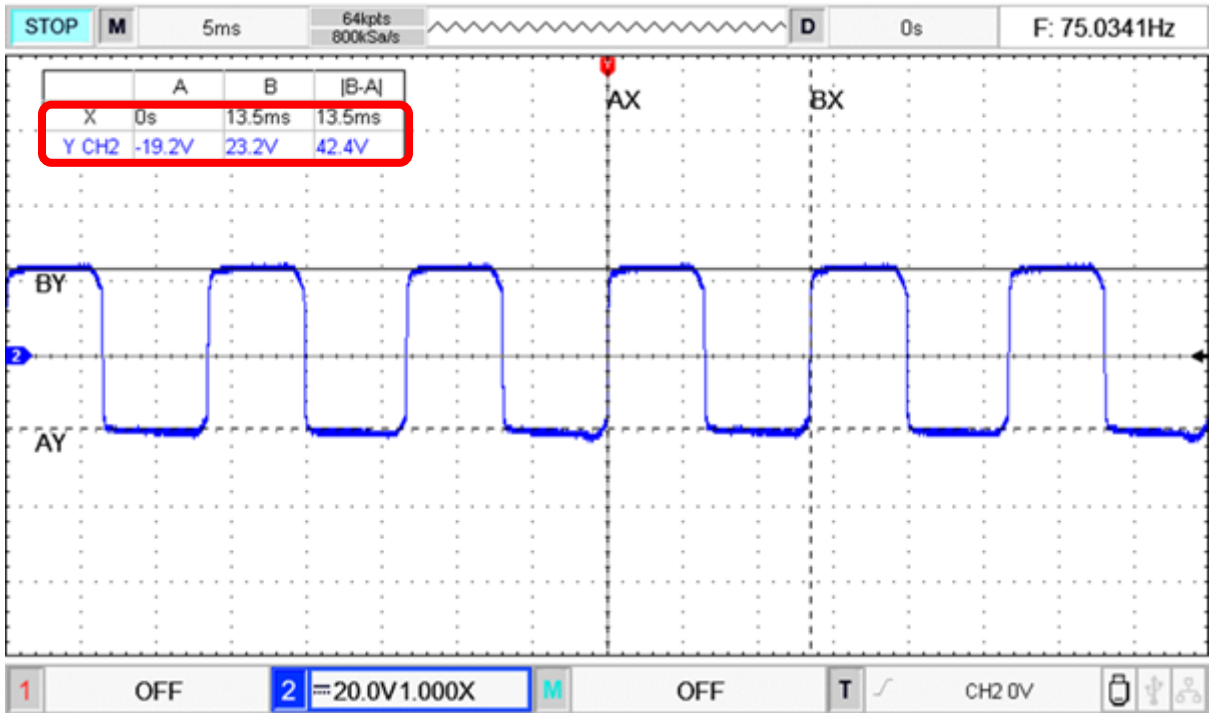


Figure 38. Loaded Waveform (Electrodes with skin contact)

Sources of Variability

- Hand fabricated electrode geometry introduces dimensional uncertainty.
- Width-to-gap ratio may vary across the electrode surface.
- Copper trace thickness and edge quality are not tightly controlled.
- Electrode-to-skin contact impedance varies with placement conditions.
- Results demonstrate proof-of-concept functionality rather than final performance characterization.

Current Monitoring and Protection Architecture

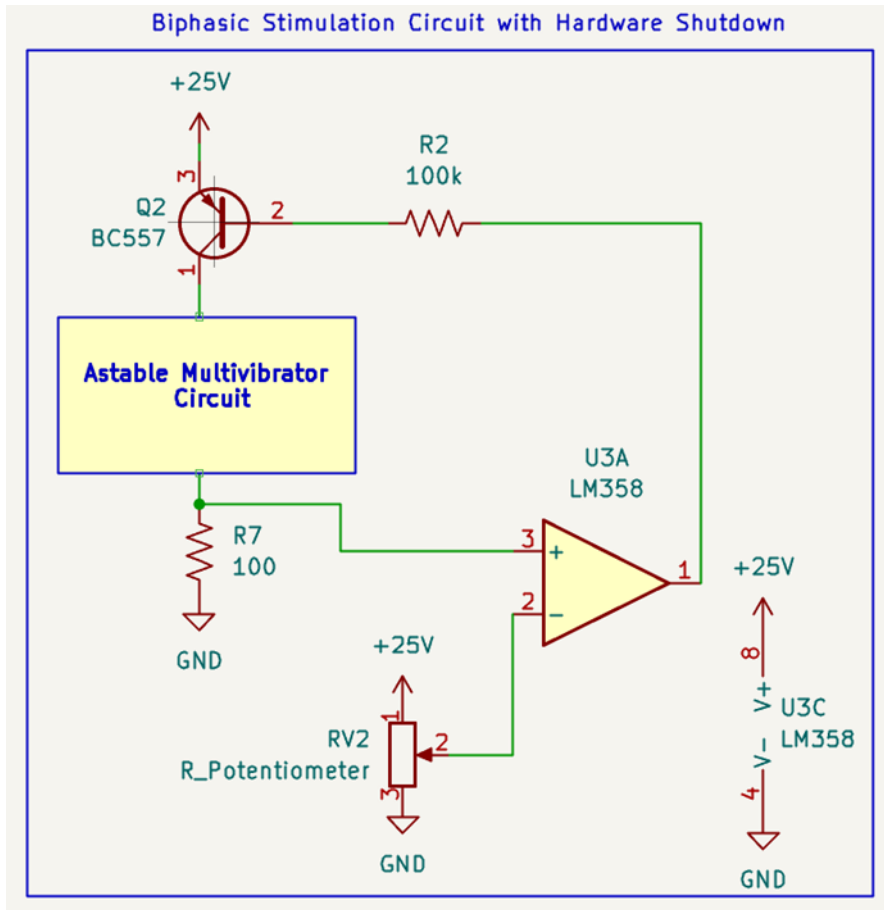


Figure 32. Schematic of the current monitoring.

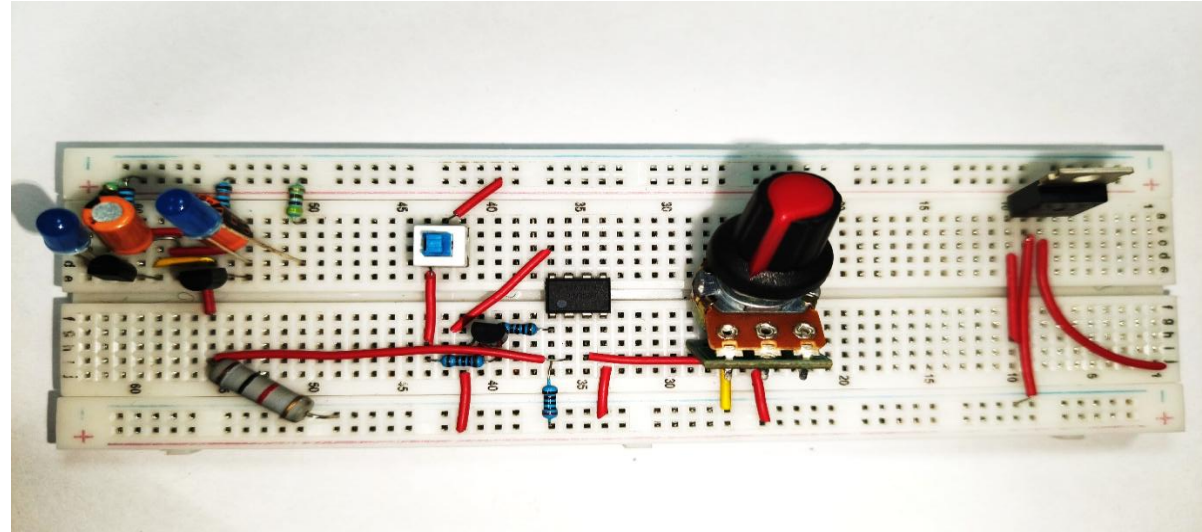
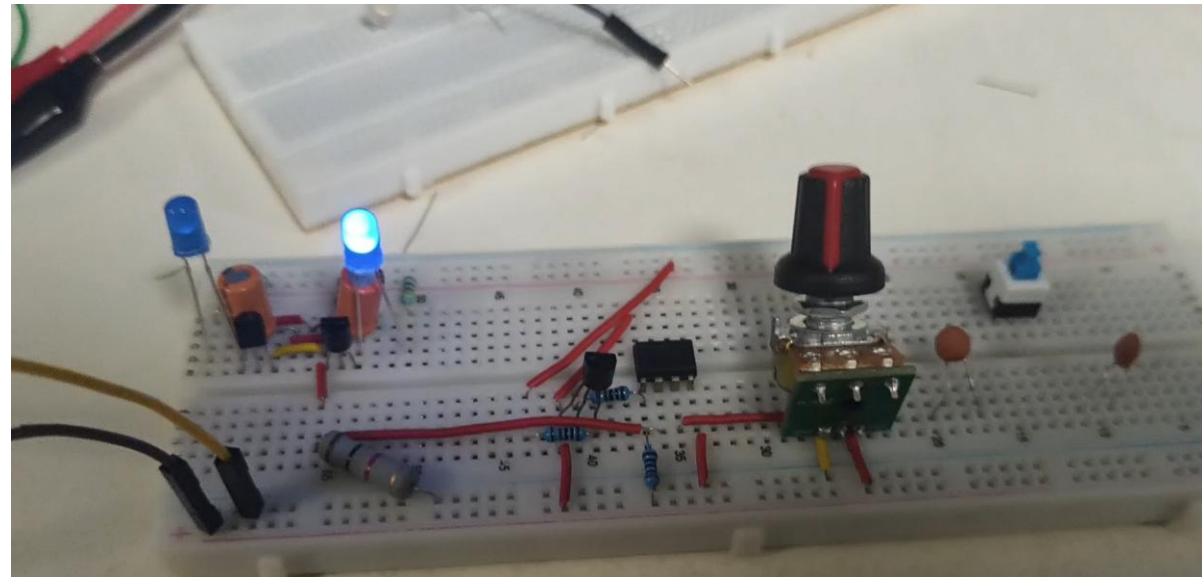


Figure 33. Breadboard implementation of current monitoring circuit.



Future Work

- Custom PCB Fabrication
- Multilayer PCB to miniaturize
- MT3608 testing
- Current monitoring integration
- Biphasic stimulation
- Safety circuitry
- Animal Studies

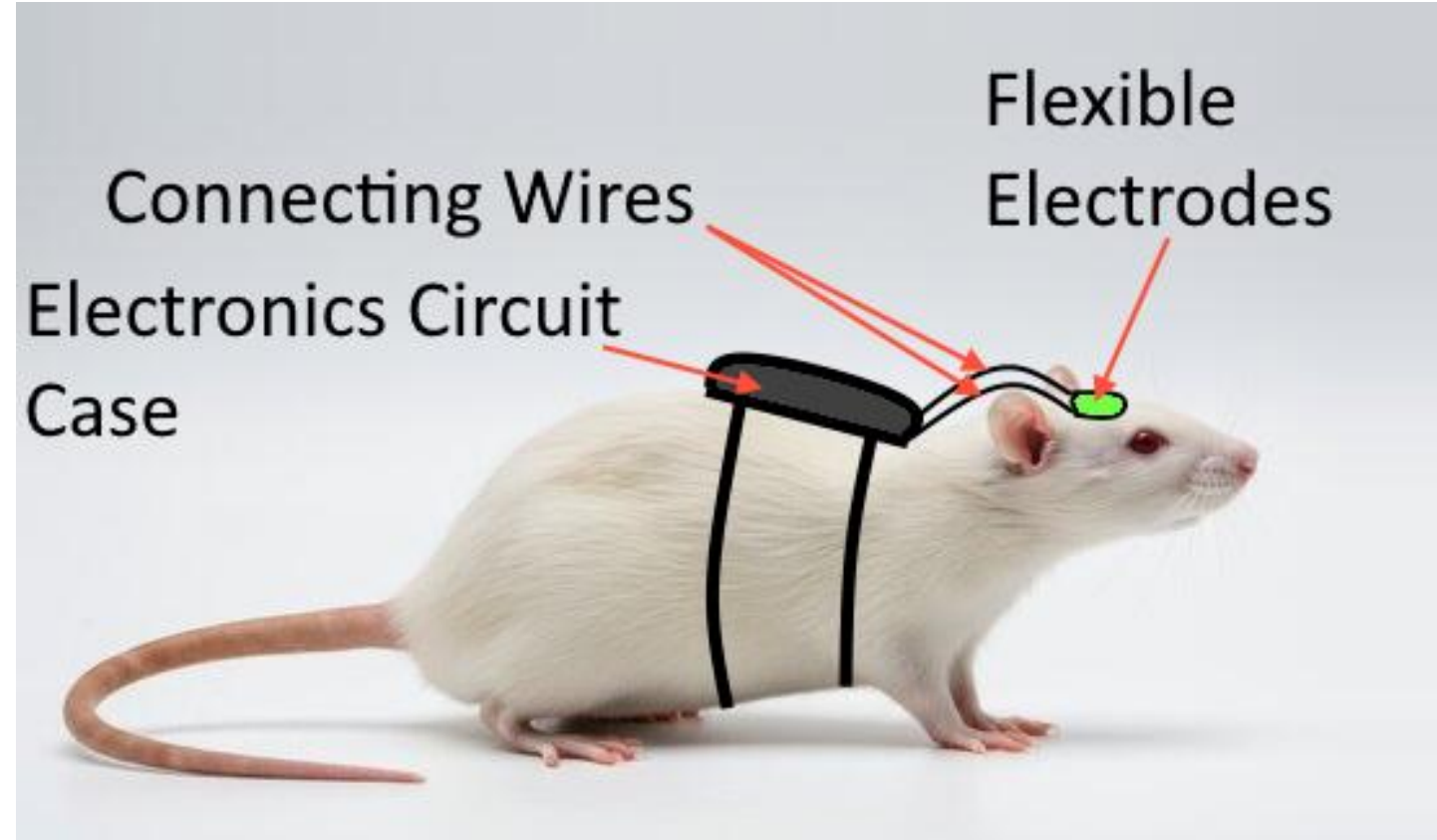
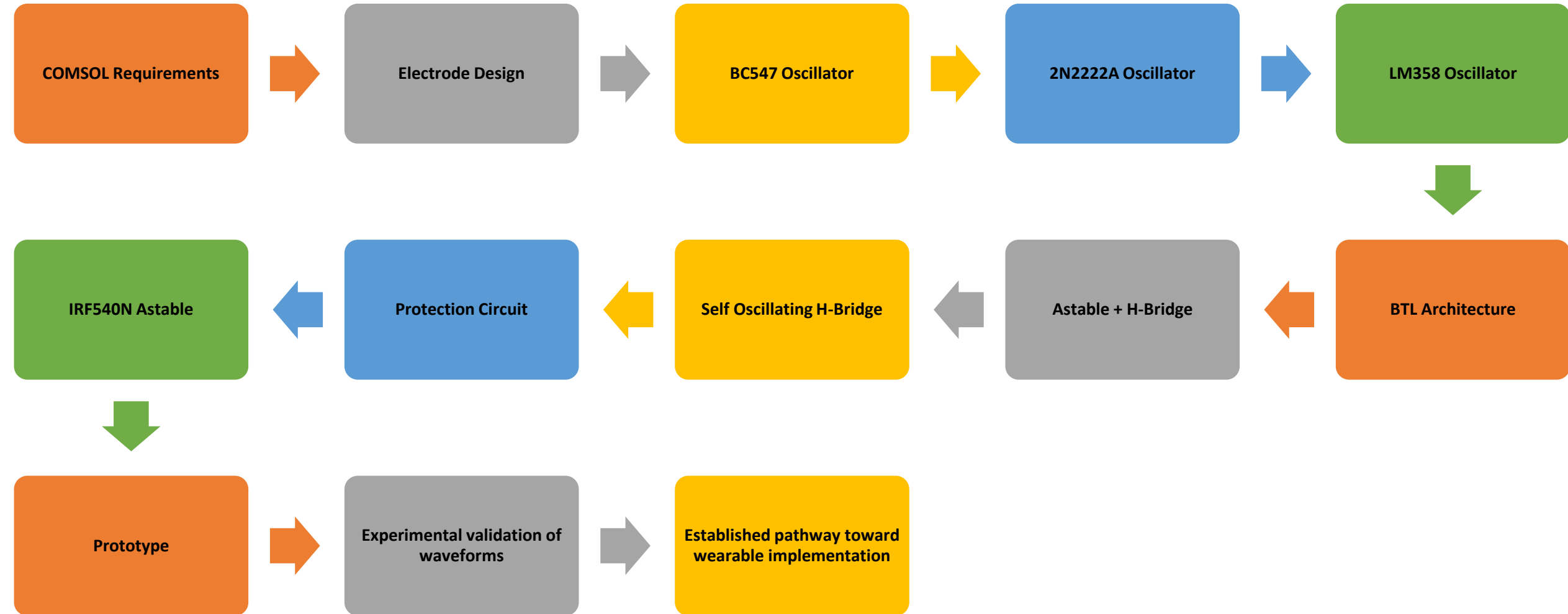


Figure 39. Proposed wearable system for small animal studies

Conclusions



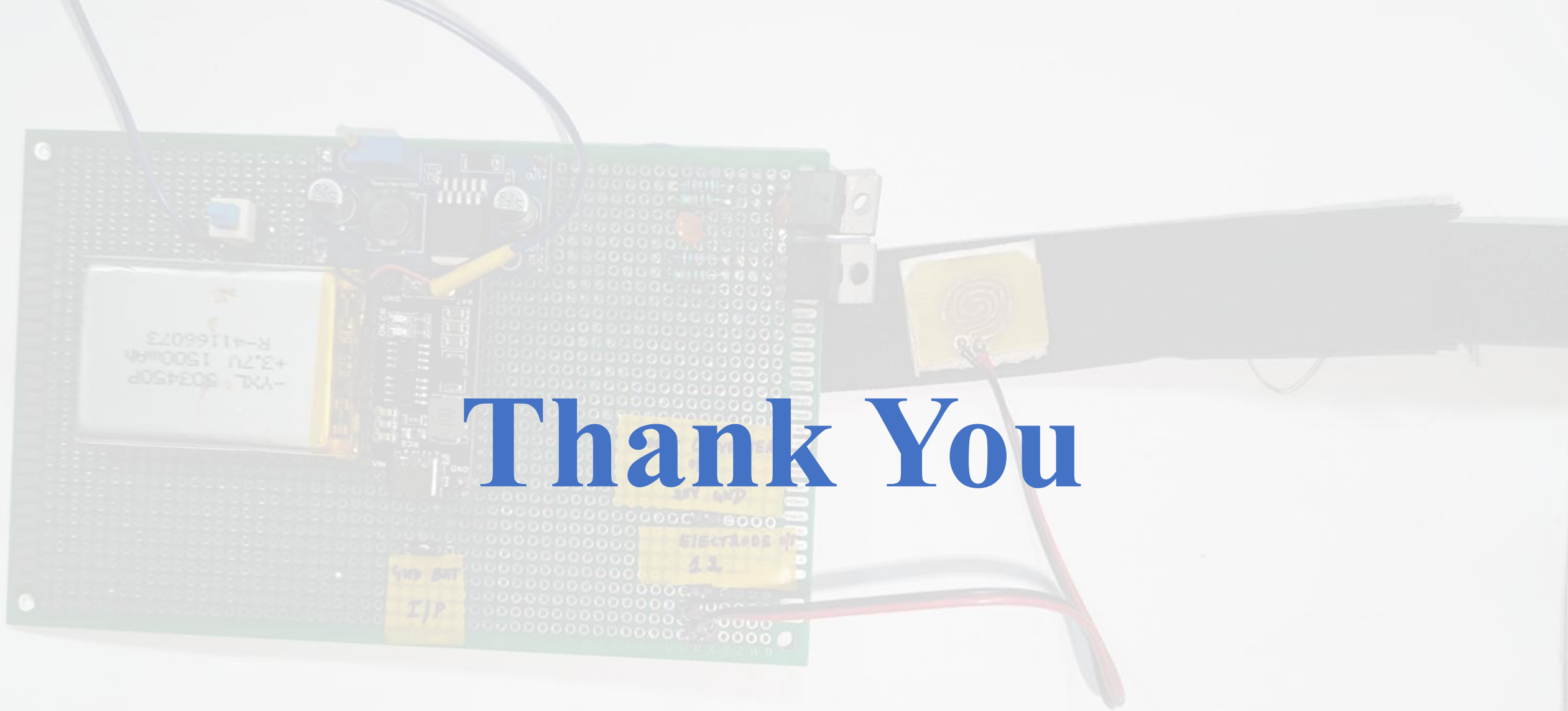
Conclusions

Each iteration addressed a specific limitation observed in the previous design

- BC547 established oscillator operation.
- 2N2222A extended operation to 25 V.
- LM358 explored comparator based generation.
- H-Bridge architectures explored direct bipolar stimulation.
- IRF540N astable multivibrator ultimately provided the best balance between performance and simplicity.

Lessons Learned from Circuit Exploration

- Simulation does not always guarantee experimental success.
- Breadboard parasitics become increasingly important at higher voltages and faster switching transitions.
- MOSFET gate capacitance significantly influence switching behavior.
- Reducing component count improved robustness and repeatability.
- Simpler architectures often provide better practical performance.
- Systematic iteration enabled identification of the most suitable stimulation topology.



Thank You